Extending the generalized Rosenblatt transform to standard Borel spaces

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Abstract

The Rosenblatt transform maps a random vector to independent uniform variables through successive conditional distribution functions; for laws on $\mathbb{R}^{\mathbb{N}}$ with atomless conditional distributions and a fixed order of the coordinates, it is the almost surely unique monotone triangular map with this property. Tests based on the transform are valid under every order, but their power can depend on this choice. We extend the generalized Rosenblatt transform to laws on countable products of standard Borel spaces, which requires, in addition to an order on the index set, a total order on each component space. For fixed orders, we establish identification and uniqueness results for the extended transform that parallel those for the classical one. We then give sufficient conditions under which the transform is invariant to each choice of order, and show when these conditions are also necessary. We also show that under every admissible order the full output retains all the statistical information in the observation about alternatives dominated by the reference, so the order affects the power of a fixed test but not the best power attainable at a given level.

1 Introduction

The Rosenblatt transform (Rosenblatt, 1952) turns a random vector with an absolutely continuous law on $\mathbb{R}^n$ into independent standard uniform variables through iterative use of the vector's conditional CDFs. When the conditional laws have atoms, the conditional CDFs jump and the output is no longer uniform. Filling each jump with an independent uniform variable allows the transform to achieve uniformity for arbitrary laws on $\mathbb{R}^n$ (Rüschendorf, 1981; Brockwell, 2007). We call this randomized construction the *generalized* Rosenblatt transform.

Rosenblatt (1952) originally proposed the transform as a preprocessing step for goodness-of-fit testing. If the conditional CDFs are taken from a hypothesized model, which we call the *reference* law, the output is uniform on $[0,1]^n$ whenever the data follow the reference, so any test of joint uniformity becomes a test

of the model. This idea underlies a range of multivariate diagnostics, historically developed for real-valued coordinates (Section 1.1). Many statistical models, however, describe data or parameters that are not real vectors, e.g. graphs, partitions, matrices, or other structured objects. Applying the transform to such a model requires a conditional CDF on each component space, and hence a total order on that space, which we call a *component order*. On $\mathbb{R}$ the usual $\leq$ order serves naturally and is generally ignored; on a space of graphs, for example, there is no such default.

We formulate an extension of the generalized Rosenblatt transform on countable products of standard Borel spaces, making explicit both the component-space orders (our new addition) and the index order, which determines which coordinates enter each conditioning set and has been noted as a relevant design choice for the transform (Rosenblatt, 1952; Berg, 2009). We call this formulation the *extended generalized Rosenblatt transform* (EGRT) and study, in three parts, how its properties depend on these order choices.

(i) *Defining and constructing the EGRT (Section 2):* We conceptualize and define the EGRT, explicitly parametrizing it by its component space and index orders. We show that the EGRT is, up to null sets, the unique uniformizing map within a class of locally randomized monotone triangular maps, mirroring the classical result (Bogachev et al., 2005).

(ii) *Order invariance (Section 3):* Given partial orders that encode the comparisons regarded as fixed, such as inclusion on a graph-valued component or the reachability order of a DAG on the index set, we ask when all of their *admissible* extensions (the total orders that extend them and satisfy the requirements in (i)) yield the same EGRT.

For component orders, with the index order fixed, the answer turns on the conditional reference mass of values incomparable with the observation, which we call *conditional ambiguity*. Its vanishing almost everywhere is sufficient for invariance, and is also necessary under mild technical conditions.

For index orders, all admissible orders give the same transform almost everywhere if and only if the reference satisfies *ancestral screening*: each coordinate is conditionally independent of the indices incomparable with it, given its ancestors. In particular, a reference that is Markov with respect to a DAG gives the same transform under all of the DAG's finite-predecessor topological orders.

(iii) *Statistical equivalence across orders (Section 4):* For each admissible order, a single measurable map recovers the observation from the output almost surely under every reference-dominated law. The original and transformed experiments are therefore equivalent over any reference-dominated family. The behavior of a fixed test can vary between orders, but we show how one test can be transferred to its equivalent test under a different order.

We characterize the output likelihood ratio as the input likelihood ratio evaluated at the reconstructed

observation. This implies that every f-divergence between the output law and uniformity equals the corresponding divergence between the input law and the reference. For dominated alternatives, we also strengthen the finite-witness result in (i): each output prefix preserves the divergence between the corresponding input and reference marginals, and these divergences increase to the full divergence as the prefixes exhaust the index set.

Proofs and supporting results are presented in the supplementary material; references to sections and results there carry the prefix S.

1.1 Related work

Soon after Rosenblatt (1952), Knothe (1957) introduced a closely related triangular map in the study of convex bodies, and the resulting Knothe–Rosenblatt rearrangement is now widely used as a transport map in sampling and density estimation (Baptista et al., 2024a,b).

The Rosenblatt transform and its generalization were both introduced on $\mathbb{R}^n$, but can be extended directly to countably many real coordinates (Bogachev et al., 2005; Rüschendorf and de Valk, 1993). For components beyond the real line, related measurable representations are known for Borel spaces: Skorokhod (1976) gives a single conditional step, Rüschendorf and de Valk (1993) note that their sequential representation extends to processes with values in Borel spaces, and Kallenberg (2021, Lemma 4.22) and Beiglböck et al. (2025, Theorem A.3) give jointly measurable representations of probability kernels, the latter for atomless conditional laws. In existing treatments of the Rosenblatt transform, an order on a component space appears, if at all, only implicitly, for example through a Borel encoding into $\mathbb{R}$ (Kechris, 1995, Theorem 15.6).

Regarding uniqueness, Bogachev et al. (2005, Lemma 2.1) shows that, for laws on $\mathbb{R}^{\mathbb{N}}$ with atomless conditionals, the Rosenblatt transform is the unique uniformizing increasing triangular transform. For laws with atoms, Backhoff et al. (2017, Proposition 5.8) characterize the Knothe–Rosenblatt coupling on finite-dimensional real spaces through bicausality and conditional monotonicity. This determines the joint law of the observation and the output, but does not specify how a given auxiliary uniform draw is used to produce the output. Our uniqueness result concerns the randomized map $(x, v) \mapsto T(x, v)$ itself.

In statistics, the transform is often used as a preprocessing step for goodness-of-fit testing. Examples include multivariate Kolmogorov–Smirnov tests (Justel et al., 1997), goodness-of-fit tests for copulas (Genest et al., 2009; Berg, 2009), and the evaluation of sequential and multivariate density forecasts (Diebold et al., 1998, 1999; Dovern and Manner, 2020). Index-order effects on fixed diagnostics have long been recognized (Rosenblatt, 1952; Rüschendorf, 2009; Berg, 2009). Dovern and Manner (2020) construct order-invariant test statistics and Pournaderi (2026) pools evidence across orders in attempts to reduce the sensitivity of

Rosenblatt-based diagnostics to the order. Closest to our index-order results, Akbari et al. (2023, Theorem 1) show that Knothe–Rosenblatt maps for compatible orders of a DAG agree up to permutation in a finite, absolutely continuous structural equation model with mechanisms strictly increasing in their independent noise arguments and strictly increasing noise CDFs. Ancestral screening is necessary as well as sufficient for index-order invariance, and its DAG consequence extends this invariance to arbitrary standard Borel components, local randomization, and countable index sets with finite ancestor sets.

When introducing the idea of orders on the component space, we start by assuming the existence of a partial order on that space. Two order-theoretic questions will arise: whether the partial order has an admissible extension, and whether its admissible extensions can detect incomparable values. Both connect to real-valued representations of preorders. An admissible extension exists exactly when some countable family of monotone Borel functions separates points, a Borel version of the injective monotones of Hack et al. (2022, Lemmas 1–3). Detecting incomparability requires a countable family of Borel functions that represents the order itself, so that x precedes y exactly when $f(x) \leq f(y)$ for every f in the family, called a countable multi-utility (Hack et al., 2022, 2023). Theorem 3.5 shows that, for a partial order with Borel graph and a nonempty extension class, such a representation exists exactly when the admissible extensions are CDF-separating for every law.

2 The extended generalized Rosenblatt transform

We begin by introducing standard results in our notation; further background can be found in Kechris (1995) and Kallenberg (2021, Chapter 1).

A measurable space $(S, \mathcal{S})$ is a *standard Borel space* if it is isomorphic to a Borel subset of a Polish space. For measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$, a *probability kernel* from S to T is a map

$$K : S \times \mathcal{T} \to [0, 1]$$

such that $K(s, \cdot)$ is a probability measure on $(T, \mathcal{T})$ for each $s \in S$, and $K(\cdot, B)$ is measurable for each $B \in \mathcal{T}$. For a measurable map $f : S \to T$ and a probability measure μ on S, we write $f_{\#}\mu$ for the pushforward measure, defined by $(f_{\#}\mu)(B) = \mu(f^{-1}(B))$.

Let $\mathcal{I}$ be a countable index set. Throughout this section we fix a total order $\trianglelefteq$ on $\mathcal{I}$ in which every index has finitely many strict predecessors, so that $(\mathcal{I}, \trianglelefteq)$ is order-isomorphic to an initial segment of $(\mathbb{N}, \leq)$. We call such an order a *finite-predecessor index order*. We write $j \triangleleft k$ for $j \trianglelefteq k$ with $j \neq k$, and call $D \subseteq \mathcal{I}$ *downward closed* if $d \in D$ and $i \trianglelefteq d$ imply $i \in D$. A finite downward-closed set will also be called a *finite*

prefix.

Let $(\mathcal{X}_k, \mathcal{B}_k)_{k\in\mathcal{I}}$ be standard Borel spaces, with product space

$$\mathcal{X}_{\mathcal{I}} := \prod_{k\in\mathcal{I}} \mathcal{X}_k, \qquad \mathcal{B}_{\mathcal{I}} := \bigotimes_{k\in\mathcal{I}} \mathcal{B}_k.$$

As a countable product of standard Borel spaces, $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}})$ is itself standard Borel (Kechris, 1995; Kallenberg, 2021). For $S \subseteq \mathcal{I}$, let $\pi_S : \mathcal{X}_{\mathcal{I}} \to \mathcal{X}_S := \prod_{j\in S} \mathcal{X}_j$ be the coordinate projection and, for a probability measure ν on $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}})$, let $\nu_S := (\pi_S)_{\#}\nu$ denote the corresponding marginal.

For $k \in \mathcal{I}$, let

$$P_{\lhd}(k) := \{j \in \mathcal{I} : j \lhd k\}.$$

For $x_{\mathcal{I}} \in \mathcal{X}_{\mathcal{I}}$, write

$$x_{\lhd k} := x_{P_{\lhd}(k)} = (x_j)_{j\lhd k}, \qquad x_{\trianglelefteq k} := (x_j)_{j\trianglelefteq k} = (x_{\lhd k}, x_k),$$

with corresponding product spaces $\mathcal{X}_{\lhd k} := \mathcal{X}_{P_{\lhd}(k)}$ and $\mathcal{X}_{\trianglelefteq k}$, product σ-algebras $\mathcal{B}_{\lhd k}$ and $\mathcal{B}_{\trianglelefteq k}$, projections $\pi_{\lhd k}$ and $\pi_{\trianglelefteq k}$, and marginals $\nu_{\lhd k}$ and $\nu_{\trianglelefteq k}$.

To define the conditional CDF at coordinate k, we condition the reference law on the preceding observations via the standard disintegration theorem (Kallenberg, 2021, Theorem 8.5).

Lemma 2.1 (Disintegration along the index order)**.** *Let ν be a probability measure on $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}})$ and fix $k \in \mathcal{I}$. There exists a $\nu_{\lhd k}$-a.e. unique probability kernel*

$$Q_k^{\nu} : \mathcal{X}_{\lhd k} \times \mathcal{B}_k \to [0, 1]$$

such that

$$\nu\{x_{\mathcal{I}} : x_{\lhd k} \in A,\ x_k \in B\} = \int_A Q_k^{\nu}(x_{\lhd k}, B)\, \nu_{\lhd k}(dx_{\lhd k})$$

for all $A \in \mathcal{B}_{\lhd k}$ and $B \in \mathcal{B}_k$. Moreover, Q_k^{ν} depends on the order $\trianglelefteq$ only through the strict predecessor set $P_{\lhd}(k)$.

We will also need conditional laws given other sets of coordinates. For $S \subseteq \mathcal{I} \setminus \{k\}$, write

$$Q_{k|S}^{\nu} : \mathcal{X}_S \times \mathcal{B}_k \to [0, 1],$$

for a fixed version of the conditional law of X_k given X_S. Such kernels exist by the same disintegration result; see also Blackwell and Dubins (1975); Kallenberg (2021).

An EGRT is defined pointwise using chosen kernel versions. Once ν and $\trianglelefteq$ are fixed, we choose one

version of each Q_k^ν. Likewise, when another conditioning set S is needed, we fix a version of $Q_{k|S}^\nu$. Let λ denote Lebesgue measure on $[0,1]$ and $\lambda^{\mathcal{I}}$ its product. For fixed orders, changing versions leaves the EGRT unchanged $\nu \otimes \lambda^{\mathcal{I}}$-a.e., and also $\mu \otimes \lambda^{\mathcal{I}}$-a.e. whenever $\mu \ll \nu$.

2.1 Borel-embeddable component orders

The EGRT progresses by iteratively turning draws from each conditional law $Q_k^\nu(x_{\lhd k}, \cdot)$ into a uniform variable. When defining the GRT on $\mathbb{R}$, the canonical total order $\leq$ is implicitly used to define CDFs. On a general standard Borel space, however, an explicit total order is needed before one can define a CDF. For our construction, we will use total orders that are Borel-embeddable in the following sense.

Definition 2.2 (Borel-embeddable total order)**.** Let $(\mathcal{X}, \mathcal{B})$ be a standard Borel space. A total order $\preceq$ on $\mathcal{X}$ is called a *Borel-embeddable total order* if there exists a Borel injection $h_\preceq : \mathcal{X} \to \mathbb{R}$ such that

$$x \preceq y \qquad \Longleftrightarrow \qquad h_\preceq(x) \leq h_\preceq(y).$$

We call any such $h_\preceq$ a *Borel order embedding* for $\preceq$.

An order embedding makes the order graph Borel, and hence makes its lower sets measurable:

$$\{(x,y) : x \preceq y\} = (h_\preceq \times h_\preceq)^{-1} \left\{(s,t) \in \mathbb{R}^2 : s \leq t\right\}.$$

Moreover, $h_\preceq$ identifies $(\mathcal{X}, \preceq)$ with an ordered Borel subset of $\mathbb{R}$, with a Borel inverse on its image (Kechris, 1995, Corollary 15.2). We write $x \prec y$ when $x \preceq y$ and $x \neq y$.

It is immediate that every standard Borel space admits such a total order by the Borel isomorphism theorem (Kechris, 1995, Theorem 15.6). However, when a component space has a canonical partial order, the analyst may want to use it, and the isomorphism theorem does not guarantee that there exists a Borel-embeddable total order which respects that partial order. We return to this question in Section 3.1.

2.2 The randomized probability integral transform

The classical probability integral transform maps a real-valued random variable X with distribution ν to $F_\nu(X)$, but if ν has atoms, this ordinary transform is not uniform. The randomized probability integral transform (rPIT) (Ferguson, 1967) resolves this by spreading the mass at each atom uniformly over the corresponding jump of the CDF. Once a Borel-embeddable order has been chosen, this construction carries over to any standard Borel space.

Fix a Borel-embeddable total order $\preceq$ on a standard Borel space $(\mathcal{X}, \mathcal{B})$. For a probability measure ν on $\mathcal{X}$, define its CDF and strict-left CDF with respect to $\preceq$ by

$$F_\nu^{\preceq}(x) := \nu\{y \in \mathcal{X} : y \preceq x\}, \qquad F_\nu^{\preceq}(x^-) := \nu\{y \in \mathcal{X} : y \prec x\}.$$

These quantities are well-defined because Borel-embeddability makes the lower sets Borel.

Definition 2.3 (Randomized probability integral transform)**.** Let ν be a probability measure on a standard Borel space $(\mathcal{X}, \mathcal{B})$ equipped with a Borel-embeddable total order $\preceq$. The *randomized probability integral transform* with respect to ν and $\preceq$ is the map

$$\mathrm{rPIT}_\nu^{\preceq} : \mathcal{X} \times [0,1] \to [0,1], \qquad \mathrm{rPIT}_\nu^{\preceq}(x, v) := F_\nu^{\preceq}(x^-) + v\, \nu(\{x\}).$$

Theorem 2.4 reproduces, at our extended level of generality, the classical result that the rPIT is uniformizing (Rüschendorf, 2009, Proposition 2.1) and adds the converse, that ν is the only input law whose rPIT is uniform.

Theorem 2.4 (Uniformity and identification for the rPIT)**.** *Let $(\mathcal{X}, \mathcal{B})$ be a standard Borel space with a Borel-embeddable total order $\preceq$, let μ and ν be probability measures on $\mathcal{X}$, and let λ be Lebesgue measure on $[0,1]$. Then*

$$(\mathrm{rPIT}_\nu^{\preceq})_{\#}(\mu \otimes \lambda) = \lambda \qquad \Longleftrightarrow \qquad \mu = \nu.$$

We also need this formula to be measurable in the conditioning value. If $(S, \mathcal{S})$ is a measurable space and $K : S \times \mathcal{B} \to [0,1]$ is a probability kernel, the map $(s, x, v) \mapsto \mathrm{rPIT}_{K(s,\cdot)}^{\preceq}(x, v)$ is jointly measurable (Lemma S2.1). This allows the one-coordinate construction to be combined sequentially to create a triangular transform.

2.3 Definition and uniformity of the EGRT

We now have the necessary components to construct the EGRT. Let $\lambda^{\mathcal{I}}$ denote product Lebesgue measure on $[0,1]^{\mathcal{I}}$. After choosing versions of the kernels $(Q_k^\nu)_{k \in \mathcal{I}}$ from Lemma 2.1, and Borel-embeddable total orders $\vec{\preceq} = (\preceq_k)_{k \in \mathcal{I}}$ on the component spaces, Lemma S2.1 shows that the coordinate functions below are measurable. Because $\mathcal{I}$ is countable, they define a measurable map

$$T_{\mathcal{I},\nu}^{\vec{\preceq},\lhd} : \mathcal{X}_{\mathcal{I}} \times [0,1]^{\mathcal{I}} \to [0,1]^{\mathcal{I}}$$

coordinatewise by

$$\big[T^{\vec{\preceq},\trianglelefteq}_{\mathcal{I},\nu}(x_{\mathcal{I}}, v_{\mathcal{I}})\big]_k := \mathrm{rPIT}^{\preceq_k}_{Q^{\nu}_k(x_{\lhd k},\cdot)}(x_k, v_k), \qquad k \in \mathcal{I}.$$

For a $\trianglelefteq$-minimal index k, the predecessor space is $\mathcal{X}_{\varnothing} = \{\varnothing\}$, and the kernel is the marginal $\nu_{\{k\}}$. This map $T^{\vec{\preceq},\trianglelefteq}_{\mathcal{I},\nu}$ is exactly our extended generalized Rosenblatt transform.

Definition 2.5 (Extended generalized Rosenblatt transform)**.** Given a reference law ν, an index set $\mathcal{I}$, an index order $\trianglelefteq$, and component-space orders $\vec{\preceq}$, the *extended generalized Rosenblatt transform* (EGRT) is the map $T^{\vec{\preceq},\trianglelefteq}_{\mathcal{I},\nu}$ above. For random elements $X_{\mathcal{I}}$ of $\mathcal{X}_{\mathcal{I}}$ and $V_{\mathcal{I}} \sim \lambda^{\mathcal{I}}$ independent of $X_{\mathcal{I}}$, we write

$$U^{\vec{\preceq},\trianglelefteq} := T^{\vec{\preceq},\trianglelefteq}_{\mathcal{I},\nu}(X_{\mathcal{I}}, V_{\mathcal{I}}),$$

and $\mathcal{L}_{\mu}(\cdot)$ for laws computed when $X_{\mathcal{I}} \sim \mu$.

When varying only one of the two order choices, we suppress the other from the notation and write $U^{\vec{\preceq}}_k$ or $U^{\trianglelefteq}_k$. We write simply U_k when both orders are understood. Like the Rosenblatt transform and GRT, the EGRT is a uniformizing transform.

Theorem 2.6 (Uniformity of the EGRT)**.** *For every probability measure ν on $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}})$ and Borel-embeddable total orders $\vec{\preceq}$ on the component spaces,*

$$(T^{\vec{\preceq},\trianglelefteq}_{\mathcal{I},\nu})_{\#}(\nu \otimes \lambda^{\mathcal{I}}) = \lambda^{\mathcal{I}}.$$

Equivalently, if $X_{\mathcal{I}} \sim \nu$, the coordinates of $U^{\vec{\preceq},\trianglelefteq}$ are i.i.d. $\mathrm{Unif}(0,1)$.

2.4 Identification and finite witnesses of misspecification

When using Rosenblatt transforms for goodness-of-fit testing (Genest et al., 2009; Berg, 2009), one specifies a reference model ν, observes data generated from an unknown law μ, and assesses whether the transformed observations behave like independent uniforms. The uniformizing property in Theorem 2.6 provides the implication

$$\mu = \nu \quad \Longrightarrow \quad (T^{\vec{\preceq},\trianglelefteq}_{\mathcal{I},\nu})_{\#}(\mu \otimes \lambda^{\mathcal{I}}) = \lambda^{\mathcal{I}}.$$

Observed non-uniformity in transformed samples therefore provides evidence against $\mu = \nu$, exactly what is required for goodness-of-fit testing. It is potentially of interest to also have the converse: if $(T^{\vec{\preceq},\trianglelefteq}_{\mathcal{I},\nu})_{\#}(\mu \otimes \lambda^{\mathcal{I}}) = \lambda^{\mathcal{I}}$, then $\mu = \nu$. For the classical Rosenblatt transform in the atomless Euclidean setting, the converse direction is implicit in the fact that the triangular transform is invertible, at least up to null sets. In the

present setting this conclusion is less immediate, but is in fact still true.

Theorem 2.7 (Identification by EGRT uniformity)**.** *Let μ and ν be probability measures on $(\mathcal{X}_\mathcal{I}, \mathcal{B}_\mathcal{I})$, and fix an index order $\trianglelefteq$ and component-space orders $\vec{\preceq}$. Then*

$$\left(T^{\vec{\preceq},\trianglelefteq}_{\mathcal{I},\nu}\right)_{\#}(\mu \otimes \lambda^\mathcal{I}) = \lambda^\mathcal{I} \qquad \Longleftrightarrow \qquad \mu = \nu.$$

In practice, goodness-of-fit tests will take in only finitely many elements at once, so we want to be sure that evidence against the hypothesis $\mu = \nu$ manifests in finitely many coordinates. Corollary 2.8 guarantees this.

Corollary 2.8 (Finite witness of misspecification)**.** *In the setting of Theorem 2.7, if $\mu \neq \nu$, then there exists a finite $\trianglelefteq$-downward closed set $D \subseteq \mathcal{I}$ such that*

$$\left(\pi_D \circ T^{\vec{\preceq},\trianglelefteq}_{\mathcal{I},\nu}\right)_{\#}(\mu \otimes \lambda^\mathcal{I}) \neq \lambda^D.$$

2.5 Uniqueness of the EGRT

A natural question is whether the EGRT is the only map satisfying Theorem 2.6. For the classical Rosenblatt transform on $\mathbb{R}^\mathbb{N}$ and a reference measure ν with atomless conditional measures, Bogachev et al. (2005) proves uniqueness after restricting to deterministic increasing triangular maps. Our setting is different in that the EGRT acts on the enlarged space $\mathcal{X}_\mathcal{I} \times [0,1]^\mathcal{I}$ and uses auxiliary randomness. The reconstruction associated with the randomized PIT recovers $X_\mathcal{I}$ from the transformed coordinates almost surely, but it does not generally recover the full auxiliary vector $V_\mathcal{I}$; in particular, atomless conditional laws make the corresponding V_k irrelevant. Thus, our setting's uniqueness statement uses a restricted notion of triangularity: the k-th output may depend on $x_{\trianglelefteq k}$ and on the current randomizer v_k, but not on earlier randomizers $v_{\triangleleft k}$.

Definition 2.9 (Locally randomized triangular map)**.** A measurable map $T : \mathcal{X}_\mathcal{I} \times [0,1]^\mathcal{I} \to [0,1]^\mathcal{I}$ is *locally randomized $\trianglelefteq$-triangular* if, for each $k \in \mathcal{I}$, there exists a measurable function

$$T_k : \mathcal{X}_{\trianglelefteq k} \times [0,1] \to [0,1]$$

such that

$$[T(x_\mathcal{I}, v_\mathcal{I})]_k = T_k(x_{\trianglelefteq k}, v_k)$$

for every $(x_\mathcal{I}, v_\mathcal{I})$.

We then define monotonicity with respect to both our observed data and the auxiliary uniform.

Definition 2.10 (Monotonicity in the current pair)**.** Let ν be a probability measure on $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}})$ and $\vec{\preceq} = (\preceq_k)_{k\in\mathcal{I}}$ be Borel-embeddable total orders on the component spaces. A locally randomized triangular map T is $\vec{\preceq}$*-monotone relative to* ν if, for every $k \in \mathcal{I}$ and $\nu_{\lhd k}$-a.e. history $x_{\lhd k}$, the map

$$(x_k, v_k) \longmapsto T_k(x_{\lhd k}, x_k, v_k)$$

is nondecreasing for the lexicographic order on $\mathcal{X}_k \times [0,1]$; that is, whenever $x_k \prec_k x'_k$, or $x_k = x'_k$ and $v_k \le v'_k$,

$$T_k(x_{\lhd k}, x_k, v_k) \le T_k(x_{\lhd k}, x'_k, v'_k).$$

For fixed orders, these two structural conditions, together with uniformization, uniquely characterize the EGRT up to null sets.

Theorem 2.11 (Uniqueness among locally randomized monotone triangular maps)**.** *Fix a probability measure ν on $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}})$ and Borel-embeddable total orders $\vec{\preceq}$ on the component spaces. The extended generalized Rosenblatt transform $T^{\vec{\preceq},\lhd}_{\mathcal{I},\nu}$ is the $\nu \otimes \lambda^{\mathcal{I}}$-a.e. unique measurable map $T : \mathcal{X}_{\mathcal{I}} \times [0,1]^{\mathcal{I}} \to [0,1]^{\mathcal{I}}$ satisfying:*

(i) T is uniformizing, $T_{\#}(\nu \otimes \lambda^{\mathcal{I}}) = \lambda^{\mathcal{I}}$;

(ii) T is locally randomized $\lhd$-triangular in the sense of Definition 2.9;

(iii) T is $\vec{\preceq}$-monotone relative to ν, in the sense of Definition 2.10.

3 Invariance under choice of order

Every EGRT constructed from the same reference law ν has output law $\lambda^{\mathcal{I}}$ under $\nu \otimes \lambda^{\mathcal{I}}$ by Theorem 2.6. This *reference invariance* is automatic by construction, but is the weakest of three levels of agreement between transforms that we use throughout the paper. For goodness-of-fit testing, it means that a test of uniformity applied to the output has the same size whichever transform is used. At the strongest level, which we call *a.e. map invariance*, two transforms agree, for $\nu \otimes \lambda^{\mathcal{I}}$-almost all values of observed data and auxiliary uniform. Thus, any diagnostics calculated over these transforms will yield identical answers. Between these two levels sits *distributional invariance*, under which the realized maps may differ in general, but their output laws coincide under every input law dominated by the reference. Every fixed test then has the same power against every reference-dominated alternative under both transforms, but they may differ on fixed data. Writing T

and T' for the two transforms,

$$\begin{aligned}
&\text{a.e. map invariance:} && T = T' \quad \nu \otimes \lambda^{\mathcal{I}}\text{-a.e.} \\
&\qquad \Downarrow \\
&\text{distributional invariance:} && T_{\#}(\mu \otimes \lambda^{\mathcal{I}}) = T'_{\#}(\mu \otimes \lambda^{\mathcal{I}}) \quad \text{for all } \mu \ll \nu \\
&\qquad \Downarrow \\
&\text{reference invariance:} && T_{\#}(\nu \otimes \lambda^{\mathcal{I}}) = T'_{\#}(\nu \otimes \lambda^{\mathcal{I}}).
\end{aligned} \tag{1}$$

The first implication holds because $\mu \ll \nu$ implies $\mu \otimes \lambda^{\mathcal{I}} \ll \nu \otimes \lambda^{\mathcal{I}}$, and the second follows by taking $\mu = \nu$. The two main results of this section, Theorems 3.10 and 3.11, characterize when the two stronger forms hold and show that, for EGRTs, a.e. map and distributional invariance are in fact equivalent.

We ask when an EGRT is essentially invariant to the choice of orders used to construct it. In goodness-of-fit testing, these orders must be chosen before any test can be run, and on spaces such as graphs there is rarely a canonical total order; invariance identifies when the choice affects neither the power of a test nor its outcome. The two order choices affect the transform in different ways, so we study their invariance properties separately. For invariance with respect to the component-space orders, we always start from a partial order on each component space, representing the comparisons the analyst regards as fixed, and the relevant decision is how to choose a total order consistent with it. Many natural spaces of interest come endowed with canonical partial orders, e.g. edge inclusion on the space of graphs on a fixed vertex set, or the Löwner order on the cone of symmetric positive definite matrices. When no such order exists and the analyst is unwilling to choose one, the trivial partial order, under which distinct elements are incomparable, still falls under our theory.

3.1 Borel-embeddable extensions of component-space partial orders

A binary relation $\preceq_p$ is a *partial order* on a set $\mathcal{X}$ if it is reflexive, antisymmetric, and transitive. A total order $\preceq$ is a *linear extension* of $\preceq_p$ if $x \preceq_p y$ implies $x \preceq y$. For a partial order on a standard Borel space, write

$$\mathcal{E}(\preceq_p) := \{\preceq : \preceq \text{ is a Borel-embeddable linear extension of } \preceq_p\}.$$

We call the elements of $\mathcal{E}(\preceq_p)$ *admissible* extensions of $\preceq_p$.

Two questions about this class matter for the rest of the section. The first is whether $\mathcal{E}(\preceq_p)$ is nonempty, which is all that constructing an EGRT requires; we address this in Proposition 3.1 and Theorem 3.2. The second is whether $\mathcal{E}(\preceq_p)$ is rich enough to detect incomparability; whenever a law puts positive mass on points left incomparable by $\preceq_p$, are there two admissible extensions that give that law different CDFs? This

richness, which we need later for Theorem 3.10, is characterized in Theorem 3.5.

Szpilrajn's extension theorem guarantees an abstract linear extension (Szpilrajn, 1930), but it does not on its own provide the Borel order embedding required by the EGRT. For countable spaces, Borel-embeddability of linear extensions is automatic.

Proposition 3.1 (Borel-embeddability on countable spaces)**.** *Let $(\mathcal{X}, \mathcal{B})$ be a countable standard Borel space and let $\preceq_p$ be a partial order on $\mathcal{X}$. Every linear extension of $\preceq_p$ is a Borel-embeddable total order on $\mathcal{X}$.*

For general standard Borel spaces, admissible extensions correspond to Borel injections into $\mathbb{R}$ that are strictly increasing along the partial order. Given such an injection h, the pullback relation $x \preceq_h y :\Leftrightarrow h(x) \leq h(y)$ is an admissible extension of $\preceq_p$, and conversely a Borel order embedding for any member of $\mathcal{E}(\preceq_p)$ is such an injection. Such an injection can in turn be built by combining any countable family of Borel functions that are monotone along $\preceq_p$ and together separate points. Hack et al. (2022, Lemmas 1–3) construct injective monotones in this way from countable separating families of increasing sets; see also Hack et al. (2023, Lemma 1(iii)) for the corresponding characterization. Together, these observations give a criterion for the extension class to be nonempty.

Theorem 3.2 (Borel-embeddable extension criterion)**.** *Let $(\mathcal{X}, \mathcal{B})$ be a standard Borel space and let $\preceq_p$ be a partial order on $\mathcal{X}$. The following are equivalent:*

(i) $\mathcal{E}(\preceq_p) \neq \varnothing$*;*

(ii) there is a Borel injection $h : \mathcal{X} \to \mathbb{R}$ such that $x \prec_p y$ implies $h(x) < h(y)$;

(iii) there is a countable family of Borel functions $f_n : \mathcal{X} \to \mathbb{R}$ that separates points and is monotone with respect to $\preceq_p$, in the sense that $x \preceq_p y$ implies $f_n(x) \leq f_n(y)$ for every n.

Given a family in (iii), enumerate $\mathbb{N} \times \mathbb{Q}$ as $((n_j, r_j))_{j \geq 1}$ and set

$$h(x) := \sum_{j=1}^{\infty} 3^{-j} \mathbf{1}\{f_{n_j}(x) > r_j\}.$$

This h satisfies (ii), and its pullback order belongs to $\mathcal{E}(\preceq_p)$.

We now move to the second question. The only values that can change sides of an observation between total orders are those left incomparable by the partial order.

Definition 3.3 (Incomparability set of a point)**.** Let $(\mathcal{X}, \mathcal{B})$ be a standard Borel space and let $\preceq_p$ be a partial order on $\mathcal{X}$. For $x \in \mathcal{X}$, define

$$I_p(x) := \{y \in \mathcal{X} : y \not\preceq_p x \text{ and } x \not\preceq_p y\}.$$

Definition 3.4 (CDF-separating extension class)**.** Let $\preceq_p$ be a partial order with Borel graph on a standard Borel space $(\mathcal{X}, \mathcal{B})$ and let $\mathcal{H}$ be a family of probability measures on $\mathcal{X}$. We call $\mathcal{E}(\preceq_p)$ *CDF-separating for* $\mathcal{H}$ if it contains a countable subfamily $\mathcal{E}_0(\preceq_p)$ such that, for every $\eta \in \mathcal{H}$ with $\eta\{x : \eta(I_p(x)) > 0\} > 0$, there exist $\preceq, \preceq' \in \mathcal{E}_0(\preceq_p)$ with

$$\eta\left\{x : F_\eta^{\preceq}(x^-) \neq F_\eta^{\preceq'}(x^-)\right\} > 0.$$

We say simply that $\mathcal{E}(\preceq_p)$ is *CDF-separating* if this holds for the family of all probability measures on $\mathcal{X}$.

The answer to the second question strengthens the condition of Theorem 3.2. Instead of only separating points, the statistics must determine the partial order, in the sense that $x \preceq_p y$ exactly when $f_n(x) \leq f_n(y)$ for every n. Such families are the countable multi-utilities of Hack et al. (2022, 2023), and Hack et al. (2022, Proposition 6) show that a countable multi-utility can be replaced by one consisting of injective monotones.

The functions $h_{n,q}$ in Theorem 3.5 are a Borel version of this replacement. Each is a Borel injective monotone, so its pullback order is admissible by Theorem 3.2, and for every incomparable pair, the family contains orders ranking them in both directions. Theorem 3.5 shows that this family is CDF-separating for every law, including laws without atoms, and that, for a partial order with a nonempty extension class, CDF separation for every law in turn requires a countable multi-utility. Countable spaces are a special case; by reflexivity and transitivity, the indicators $\mathbf{1}\{a \preceq_p \cdot\}$, $a \in \mathcal{X}$, form a countable multi-utility (see also Lemma S3.2).

Theorem 3.5 (CDF separation and countable order representations)**.** *Let $(\mathcal{X}, \mathcal{B})$ be a standard Borel space with partial order $\preceq_p$. The following are equivalent:*

(i) there are Borel functions $f_n : \mathcal{X} \to \mathbb{R}$, $n \geq 1$, such that

$$x \preceq_p y \quad \Longleftrightarrow \quad f_n(x) \leq f_n(y) \text{ for every } n \geq 1; \tag{2}$$

(ii) $\preceq_p$ has Borel graph and $\mathcal{E}(\preceq_p)$ is nonempty and CDF-separating for all probability measures on $\mathcal{X}$.

Given a family in (i), choose a Borel injection $h_0 : \mathcal{X} \to (0,1)$ that is strictly increasing along $\prec_p$, as supplied by Theorem 3.2. A countable separating family consists of the pullback orders of

$$h_{n,q}(x) := \mathbf{1}\{f_n(x) > q\} + h_0(x), \qquad n \geq 1,\ q \in \mathbb{Q}.$$

Example 3.6 (Graph inclusion)**.** Let $\mathcal{X} = \{0,1\}^E$ be the space of simple graphs on a fixed countably infinite labelled vertex set, where E is the countable set of possible edges. Equip $\mathcal{X}$ with its product Borel σ-algebra

and order it by edge inclusion. The coordinate functions $f_e(G) := \mathbf{1}\{e \in G\}$ are Borel and satisfy

$$G \subseteq H \quad \Longleftrightarrow \quad f_e(G) \le f_e(H) \text{ for every } e \in E.$$

By Theorem 3.5, the class of admissible extensions is therefore nonempty and CDF-separating for every probability law on this uncountable graph space.

3.2 Exact invariance across component-space extensions

Section 3.1 establishes when the class of admissible extensions of a partial order is nonempty, and when it is CDF-separating. An admissible extension is typically not unique, so it is of interest to know how much the choice matters. We start by setting up the building block of this analysis, *conditional ambiguity*.

Fix a finite-predecessor index order $\trianglelefteq$ and the reference kernels $(Q_k^\nu)_{k\in\mathcal{I}}$. For each k, suppose that $\mathcal{X}_k$ carries a partial order $\preceq_{p,k}$ with Borel graph and a nonempty class $\mathcal{E}(\preceq_{p,k})$ of admissible extensions. Write $I_{p,k}(x)$ for the incomparability set of Definition 3.3 computed in $(\mathcal{X}_k, \preceq_{p,k})$.

Two admissible extensions of $\preceq_{p,k}$ can disagree about whether a point lies below or above x_k only if that point is incomparable to x_k and thus in $I_{p,k}(x_k)$. Therefore, given the predecessor history $x_{\triangleleft k}$, the reference conditional probability of this set, which we call the *conditional ambiguity*, bounds how much the choice of extension can matter at x_k.

Definition 3.7 (Conditional ambiguity)**.** For a probability measure ν, the *conditional ambiguity* at coordinate k is

$$\Delta_k(x_{\trianglelefteq k}) := Q_k^\nu\big(x_{\triangleleft k}, I_{p,k}(x_k)\big).$$

This map is $\mathcal{B}_{\trianglelefteq k}$-measurable, so a.e. statements about it are well posed; see Lemma S3.1.

For a graph-valued component ordered by inclusion as in Example 3.6, Δ_k is the reference conditional probability of a graph that neither contains the observed graph nor is contained in it.

Because conditional ambiguity bounds how far the choice of extension can move each coordinate of the transform, ensuring that the ambiguity is 0 forces every admissible choice to produce the same transform.

Proposition 3.8 (Ambiguity bounds the effect of the component order)**.** *For every pair of extension sequences* $\vec{\preceq}, \vec{\preceq}' \in \prod_{k\in\mathcal{I}} \mathcal{E}(\preceq_{p,k})$*, every* $k \in \mathcal{I}$*, and every* $(x, v) \in \mathcal{X}_\mathcal{I} \times [0,1]^\mathcal{I}$*,*

$$\big|[T^{\vec{\preceq}}_{\mathcal{I},\nu}(x,v)]_k - [T^{\vec{\preceq}'}_{\mathcal{I},\nu}(x,v)]_k\big| \le \Delta_k(x_{\trianglelefteq k}).$$

Consequently, if $\Delta_k = 0$ $\nu_{\trianglelefteq k}$*-a.e. for every* k*, then* $T^{\vec{\preceq}}_{\mathcal{I},\nu} = T^{\vec{\preceq}'}_{\mathcal{I},\nu}$ $\mu \otimes \lambda^\mathcal{I}$*-a.e. for every* $\mu \ll \nu$*, including* $\mu = \nu$*,*

and the two transforms have the same output law under every such μ.

Example 3.9 (Ambiguity on a space of graphs)**.** Consider a single graph-valued component on a fixed vertex set, with two possible edges e_1, e_2. Identifying each graph with its edge set, the component space consists of

$$H_0 = \varnothing, \qquad H_1 = \{e_1\}, \qquad H_2 = \{e_2\}, \qquad H_{12} = \{e_1, e_2\}.$$

Order these graphs by inclusion and let ν be uniform on the four graphs. The only incomparable pair is (H_1, H_2), so the ambiguity is $\Delta(H_1) = \Delta(H_2) = 1/4$, while $\Delta(H_0) = \Delta(H_{12}) = 0$. The two admissible extensions are

$$\preceq : H_0 \prec H_1 \prec H_2 \prec H_{12},$$
$$\preceq' : H_0 \prec' H_2 \prec' H_1 \prec' H_{12}.$$

At the observation H_1, their transforms are

$$\mathrm{rPIT}_\nu^{\preceq}(H_1, v) = \frac{1+v}{4}, \qquad \mathrm{rPIT}_\nu^{\preceq'}(H_1, v) = \frac{2+v}{4}.$$

Changing the order moves the transformed value by exactly $1/4$, attaining the ambiguity bound in Proposition 3.8. The transform at H_2 moves by the same amount in the opposite direction, while the transforms at H_0 and H_{12} are unchanged.

For the converse, that invariance implies zero ambiguity, the class of extensions must be rich enough to reveal positive ambiguity, which is why we need Definition 3.4. At coordinate k the relevant family is

$$\mathcal{H}_k^\nu := \{Q_k^\nu(x_{\lhd k}, \cdot) : x_{\lhd k} \in \mathcal{X}_{\lhd k}\}, \tag{3}$$

the conditional laws that the EGRT actually uses at k. The class $\mathcal{E}(\preceq_{p,k})$ is CDF-separating for $\mathcal{H}_k^\nu$, whatever the reference kernels, when $\mathcal{X}_k$ is countable (Lemma S3.2) or, more generally, satisfies Equation (2) (Theorem 3.5).

Theorem 3.10 (Exact component-order invariance)**.** *Assume that each $\preceq_{p,k}$ has Borel graph and that $\mathcal{E}(\preceq_{p,k})$ is CDF-separating for $\mathcal{H}_k^\nu$ of Equation* (3)*, for every $k \in \mathcal{I}$. The following are equivalent:*

(i) ***zero conditional ambiguity a.e.:*** *for every $k \in \mathcal{I}$,*

$$\Delta_k = 0 \qquad \nu_{\trianglelefteq k}\text{-a.e.};$$

(ii) ***a.e. map invariance:*** *for every pair of extension sequences* $\vec{\preceq}, \vec{\preceq}' \in \prod_{k\in\mathcal{I}} \mathcal{E}(\preceq_{p,k})$*,*

$$T^{\vec{\preceq}}_{\mathcal{I},\nu}(x,v) = T^{\vec{\preceq}'}_{\mathcal{I},\nu}(x,v) \qquad \nu\otimes\lambda^{\mathcal{I}}\text{-a.e.};$$

(iii) ***distributional invariance under reference-dominated measures:*** *for every* $\mu \ll \nu$ *and pair of extension sequences,*

$$(T^{\vec{\preceq}}_{\mathcal{I},\nu})_{\#}(\mu\otimes\lambda^{\mathcal{I}}) = (T^{\vec{\preceq}'}_{\mathcal{I},\nu})_{\#}(\mu\otimes\lambda^{\mathcal{I}}).$$

Under any of these conditions, the equality in (ii) also holds $\mu\otimes\lambda^{\mathcal{I}}$*-a.e. for every* $\mu \ll \nu$*.*

Proposition 3.8 gives (i)⇒(ii), and the hierarchy Equation (1) gives (ii)⇒(iii), both without requiring CDF separation; CDF separation is used only for (iii)⇒(i).

3.3 Exact invariance across index-order extensions

We now fix Borel-embeddable component-space orders $\vec{\preceq} = (\preceq_k)_{k\in\mathcal{I}}$ and vary the order on the index set. A component-order extension changes which values of a fixed conditional distribution count as lower than the observation, while an index order changes the conditioning set itself: under a finite-predecessor index order $\trianglelefteq$, the kernel at coordinate k is $Q^{\nu}_{k} = Q^{\nu}_{k|P_{\trianglelefteq}(k)}$. Exact invariance must consequently come from a probabilistic reduction of these conditional kernels. We write $T^{\trianglelefteq}_{\mathcal{I},\nu}$ when the fixed component orders are suppressed.

Let $\trianglelefteq_p$ be a partial order on $\mathcal{I}$. For $k \in \mathcal{I}$, define its strict ancestors and incomparable indices by

$$\mathrm{anc}_p(k) := \{j : j \triangleleft_p k\}, \qquad \mathrm{inc}_p(k) := \{j \in \mathcal{I}\setminus\{k\} : j \not\trianglelefteq_p k \text{ and } k \not\trianglelefteq_p j\}.$$

Let $\mathcal{L}(\trianglelefteq_p)$ be the set of finite-predecessor linear extensions of $\trianglelefteq_p$; we call its elements *admissible* index orders. If every ancestor set is finite, this class is nonempty (Lemma S3.3). For every $\trianglelefteq \in \mathcal{L}(\trianglelefteq_p)$,

$$\mathrm{anc}_p(k) \subseteq P_{\trianglelefteq}(k) \subseteq \mathrm{anc}_p(k) \cup \mathrm{inc}_p(k).$$

We now have what we need for the index-order counterpart of Theorem 3.10. An admissible index order cannot change whether an ancestor of k precedes k: every element of $\mathrm{anc}_p(k)$ belongs to $P_{\trianglelefteq}(k)$ under every $\trianglelefteq \in \mathcal{L}(\trianglelefteq_p)$. What may change is which finite subset of the indices incomparable with k is also placed before k.

Theorem 3.11 (Exact index-order invariance)**.** *Let* $\trianglelefteq_p$ *be a partial order on the countable set* $\mathcal{I}$*, with finite* $\mathrm{anc}_p(k)$ *for every* $k \in \mathcal{I}$*, and* ν *be a probability measure on* $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}})$*. The following are equivalent:*

(i) ***ancestral screening of incomparable indices:*** *for every* $k \in \mathcal{I}$,

$$X_k \perp\!\!\!\perp_\nu X_{\mathrm{inc}_p(k)} \mid X_{\mathrm{anc}_p(k)};$$

(ii) ***ancestor-kernel reduction:*** *for every* $k \in \mathcal{I}$ *and every* $\trianglelefteq \in \mathcal{L}(\trianglelefteq_p)$,

$$Q^\nu_{k|P_\trianglelefteq(k)}\big(X_{P_\trianglelefteq(k)}, \cdot\big) = Q^\nu_{k|\mathrm{anc}_p(k)}\big(X_{\mathrm{anc}_p(k)}, \cdot\big) \qquad \nu\text{-a.e.};$$

(iii) ***a.e. map invariance:*** *for every pair* $\trianglelefteq, \trianglelefteq' \in \mathcal{L}(\trianglelefteq_p)$,

$$T^{\trianglelefteq}_{\mathcal{I},\nu}(x, v) = T^{\trianglelefteq'}_{\mathcal{I},\nu}(x, v) \qquad \nu \otimes \lambda^{\mathcal{I}}\text{-a.e.};$$

(iv) ***distributional invariance under reference-dominated measures:*** *for every* $\mu \ll \nu$ *and every pair* $\trianglelefteq, \trianglelefteq' \in \mathcal{L}(\trianglelefteq_p)$,

$$\big(T^{\trianglelefteq}_{\mathcal{I},\nu}\big)_{\#}(\mu \otimes \lambda^{\mathcal{I}}) = \big(T^{\trianglelefteq'}_{\mathcal{I},\nu}\big)_{\#}(\mu \otimes \lambda^{\mathcal{I}}).$$

Under any of these conditions, the equality in (iii) *also holds* $\mu \otimes \lambda^{\mathcal{I}}$*-a.e. for every* $\mu \ll \nu$.

Ancestral screening has a familiar interpretation through the local Markov property for Bayesian networks (Lauritzen, 1996). Let $G = (\mathcal{I}, E)$ be a DAG on a countable vertex set, and let its *reachability relation* be

$$j \trianglelefteq_p k \quad :\Longleftrightarrow \quad j = k, \text{ or there is a directed path from } j \text{ to } k \text{ in } G.$$

Acyclicity makes this a partial order such that its ancestors are the ordinary graph ancestors, and its incomparable indices are the non-ancestor non-descendants. Write $\mathrm{Pa}_G(k) := \{j : (j, k) \in E\}$ for the parents of k, call j a *descendant* of k when $k \triangleleft_p j$, and write $\mathrm{ND}_G(k) := \mathcal{I} \setminus (\{k\} \cup \{j : k \triangleleft_p j\})$ for its non-descendants. The non-descendants decompose as the disjoint union $\mathrm{ND}_G(k) = \mathrm{anc}_p(k) \cup \mathrm{inc}_p(k)$. We say that ν is *Markov with respect to* G if it satisfies the local Markov property

$$X_k \perp\!\!\!\perp_\nu X_{\mathrm{ND}_G(k)\setminus \mathrm{Pa}_G(k)} \mid X_{\mathrm{Pa}_G(k)}, \qquad k \in \mathcal{I}.$$

Corollary 3.12 (EGRT invariance across topological orders)**.** *Let* G *be a DAG on the countable set* $\mathcal{I}$ *with every* $\mathrm{anc}_p(k)$ *finite, and suppose* ν *is Markov with respect to* G*. Then ancestral screening, condition* (i) *of Theorem 3.11, holds for the reachability order* $\trianglelefteq_p$ *of* G*. Consequently every pair of finite-predecessor topological orders of* G *produces the same EGRT* $\nu \otimes \lambda^{\mathcal{I}}$*-a.e., as well as the same output law under every*

$\mu \ll \nu$. *Its k-th coordinate is*

$$\mathrm{rPIT}^{\preceq_k}_{Q^{\nu}_{k|\mathrm{Pa}_G(k)}(x_{\mathrm{Pa}_G(k)},\cdot)}(x_k, v_k) \qquad \nu \otimes \lambda^{\mathcal{I}}\text{-a.e.}$$

4 Statistical equivalence across orders

Section 3 characterizes when the choice of orders leaves the EGRT unchanged. When those conditions fail, the EGRT can produce different output laws for different orders, leading to differences in the power of a fixed test. This raises the concern that choosing a particular set of orders could entirely preclude the analyst from observing some deviation from the reference law after transformation. We show that this does not happen; regardless of the choice of order, no information about the data is lost, provided the data follow a law dominated by the reference. This, in turn, implies that the available information after transformation is identical for any two choices of orders; in fact, we give a construction for how to convert any test designed for one order into its equivalent test under another.

While every test has an equivalent test under any other order, a fixed test will generally differ between two orders. We show that changing the order moves the evidence against the reference around the output space without changing its overall amount. For a dominated law, the output law has a density with respect to the uniform law, namely the likelihood ratio of the data law to the reference evaluated at the observation reconstructed from the output. Its distribution under the uniform law is the same for every order, so every f-divergence between the output law and uniformity equals the corresponding divergence between the data law and the reference. Differences in the power of a fixed test therefore are a result of where the evidence appears in the output, rather than the overall amount of evidence.

In practice, a test can examine only finitely many output coordinates. Corollary 2.8 shows that every departure from the reference is visible in some finite prefix of the output, but doesn't capture *how* visible. We show that the f-divergence result above holds for arbitrary finite prefixes as well, so restricting the output to a prefix loses exactly as much information as restricting the data to those coordinates. As the prefixes grow to include every coordinate, the full f-divergence is recovered.

4.1 Reconstruction, equivalence, and test transfer

Fix the reference law ν and a nonempty class $\mathfrak{O}$ of admissible order specifications $o = (\vec{\preceq}, \trianglelefteq)$: each o chooses Borel-embeddable total orders on the component spaces and a finite-predecessor index order. Write T_o for the EGRT under o and

$$\rho^{\mu}_o := (T_o)_{\#}(\mu \otimes \lambda^{\mathcal{I}}),$$

for its output law under μ, so that $\rho_o^\nu = \lambda^{\mathcal{I}}$ by Theorem 2.6. We first establish that the original observation can be recovered from this output alone.

Lemma 4.1 (Measurable reconstruction from an EGRT output)**.** *Fix $o \in \mathfrak{O}$. There exists a measurable reconstruction map, depending only on ν and o,*

$$G_o : [0,1]^{\mathcal{I}} \to \mathcal{X}_{\mathcal{I}}$$

such that, for every $\mu \ll \nu$,

$$G_o(T_o(X,V)) = X \qquad \mu \otimes \lambda^{\mathcal{I}}\text{-a.e.}, \tag{4}$$

and hence $(G_o)_{\#}\rho_o^\mu = \mu$.

This reconstruction also gives equivalence of the original and transformed statistical experiments (Blackwell, 1953; Torgersen, 1991). An *experiment* is a family of probability measures on a common measurable space, indexed by the unknown input law $\mu \in \mathcal{M}$. Two experiments are called *equivalent* if each can be obtained from the other via a μ-independent Markov kernel, in which case they share the same risk functions for all randomized decision rules (and, in particular, the same power functions for tests). Here the forward and reconstruction kernels are

$$\mathsf{P}_o(x,B) := \int_{[0,1]^{\mathcal{I}}} \mathbf{1}\{T_o(x,v) \in B\}\ \lambda^{\mathcal{I}}(\mathrm{d}v),$$

$$\mathsf{R}_o(u,A) := \mathbf{1}\{G_o(u) \in A\},$$

for Borel sets $B \subseteq [0,1]^{\mathcal{I}}$ and $A \subseteq \mathcal{X}_{\mathcal{I}}$. Writing $(\mu\mathsf{P}_o)(B) := \int \mathsf{P}_o(x,B)\,\mu(\mathrm{d}x)$, we have $\mu\mathsf{P}_o = \rho_o^\mu$, while reconstruction gives

$$(\mu\mathsf{P}_o)\mathsf{R}_o = \mu \qquad \text{for every } \mu \ll \nu. \tag{5}$$

In particular, $(G_o)_{\#}\lambda^{\mathcal{I}} = \nu$.

Theorem 4.2 (Equivalence of the original and transformed experiments)**.** *Let $\mathcal{M}$ be any family of probability measures on $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}})$ such that $\mu \ll \nu$ for every $\mu \in \mathcal{M}$, and let $o, o' \in \mathfrak{O}$. Then the original experiment $(\mu)_{\mu\in\mathcal{M}}$ and the transformed experiments $(\rho_o^\mu)_{\mu\in\mathcal{M}}$ and $(\rho_{o'}^\mu)_{\mu\in\mathcal{M}}$ are equivalent.*

Equivalence of experiments suggests that a test under one experiment has a corresponding test with the same behavior under the other experiment. We can construct the mapping between tests explicitly. A test ψ on the output under o, applied after transforming $X = x$, rejects with conditional probability

$$(\mathsf{P}_o\psi)(x) := \int_{[0,1]^{\mathcal{I}}} \psi(u)\,\mathsf{P}_o(x,\mathrm{d}u) = \int_{[0,1]^{\mathcal{I}}} \psi\big(T_o(x,v)\big)\,\lambda^{\mathcal{I}}(\mathrm{d}v).$$

Reconstructing x from the output under o' therefore gives the following test on that output.

Corollary 4.3 (Transfer of tests between orders)**.** *Fix $o, o' \in \mathfrak{O}$ and a measurable test $\psi : [0,1]^{\mathcal{I}} \to [0,1]$ for the output under o. Define its transfer to order o' by*

$$\psi'(u') := (\mathsf{P}_o\psi)(G_{o'}(u')) = \int_{[0,1]^{\mathcal{I}}} \psi(T_o(G_{o'}(u'), v))\ \lambda^{\mathcal{I}}(\mathrm{d}v). \tag{6}$$

Then ψ' is a measurable test with values in $[0,1]$. For every $\mu \ll \nu$,

$$\int \psi'\, \mathrm{d}\rho^{\mu}_{o'} = \int \psi\, \mathrm{d}\rho^{\mu}_{o}. \tag{7}$$

To implement the transferred test, reconstruct $x = G_{o'}(u')$, draw fresh independent auxiliary uniforms, apply T_o to x, and run ψ on the resulting output. This procedure has conditional rejection probability $\psi'(u')$ and avoids evaluating the integral in Equation (6).

Example 4.4 (A tail test with a disconnected transfer)**.** Consider a single graph-valued component on a fixed vertex set, with $m \geq 2$ possible edges $e_1, \ldots, e_m$, and write $z_j(H) := \mathbf{1}\{e_j \in H\}$. Under the reference law ν, let the edge indicators be independent, with $\nu\{z_1 = 1\} = \alpha \in (0,1)$ and $\nu\{z_j = 1\} = 1/2$ for $j \geq 2$. Put $N := 2^{m-1}$ and let $B(H) := \sum_{j=2}^{m} 2^{m-j} z_j(H)$ be the binary rank of the remaining edges. Compare the orders o, o' obtained by sorting the graphs by increasing values of

$$h_o(H) := N z_1(H) + B(H), \qquad h_{o'}(H) := 2B(H) + z_1(H).$$

Both orders extend graph inclusion, but the first prioritizes the presence of e_1, whereas the second prioritizes every other edge.

Under o, the graphs containing e_1 occupy the output interval $(1-\alpha, 1)$. Thus the upper-tail test $\psi(u) = \mathbf{1}\{u > 1-\alpha\}$ has size α and rejects exactly when e_1 is present, almost surely. Under o', each value b of B occupies a block of length $1/N$, and the graphs containing e_1 occupy the final α fraction of each block. The transfer in Equation (6) is therefore

$$\psi'(u') = \mathbf{1}\left\{ u' \in \bigcup_{b=0}^{N-1} \left(\frac{b+1-\alpha}{N}, \frac{b+1}{N} \right) \right\},$$

up to a Lebesgue-null set. Under every law η on this graph space, ψ applied under o and ψ' applied under o' both reject with probability $\eta\{z_1 = 1\}$.

4.2 Likelihood ratios and divergence preservation

The output likelihood ratio describes how the evidence appears under a particular order: it is the input likelihood ratio evaluated at the reconstructed observation.

Throughout this subsection, fix $o \in \mathfrak{O}$ and $\mu \ll \nu$, and write $R := \mathrm{d}\mu/\mathrm{d}\nu$ for the input likelihood ratio.

Proposition 4.5 (Output likelihood ratio)**.** *We have $\rho_o^\mu \ll \lambda^{\mathcal{I}}$ and*

$$\frac{\mathrm{d}\rho_o^\mu}{\mathrm{d}\lambda^{\mathcal{I}}}(u) = R(G_o(u)) \qquad \lambda^{\mathcal{I}}\text{-a.e.} \tag{8}$$

Since $(G_o)_\# \lambda^{\mathcal{I}} = \nu$, the distribution of $R \circ G_o$ under $\lambda^{\mathcal{I}}$ is the distribution of R under ν. Although changing the order alters the output likelihood ratio as a pointwise function of u, its distribution under the uniform law remains invariant.

This also preserves every f-divergence (Ali and Silvey, 1966; Csiszár, 1967). For probability measures $P \ll Q$ and a convex function $f : (0, \infty) \to \mathbb{R}$ with $f(1) = 0$, extend f to zero by $f(0) := \lim_{t \downarrow 0} f(t) \in \mathbb{R} \cup \{+\infty\}$ and define

$$D_f(P\|Q) := \int f\left(\frac{\mathrm{d}P}{\mathrm{d}Q}\right) \mathrm{d}Q.$$

An f-divergence thus depends only on the distribution of the likelihood ratio under the reference measure Q.

Corollary 4.6 (Preservation of divergences)**.** *For every f as above,*

$$D_f\big(\rho_o^\mu \big\| \lambda^{\mathcal{I}}\big) = D_f(\mu\|\nu), \tag{9}$$

including when both sides are infinite.

The equality also follows from the data-processing inequality for f-divergences (Liese, 2012, Theorem 3.1), applied to the forward and reconstruction kernels. Taking $f(t) = |t-1|/2$, $t \log t$, $(\sqrt{t}-1)^2$, or $(t-1)^2$ gives preservation of total variation, Kullback–Leibler divergence, squared Hellinger distance (without a factor $1/2$), or χ^2 divergence, respectively.

These equalities provide a quantitative counterpart to the information preservation of Section 4.1, though a fixed rejection region in the uniform space may still have order-dependent probability (as in Example 4.4) because $u \mapsto R(G_o(u))$ varies with G_o.

For the alternative $\mu = \nu(\cdot \mid z_1 = 1)$ in Example 4.4, the two output densities are

$$\frac{\mathrm{d}\rho_o^\mu}{\mathrm{d}\lambda}(u) = \frac{1}{\alpha}\psi(u), \qquad \frac{\mathrm{d}\rho_{o'}^\mu}{\mathrm{d}\lambda}(u) = \frac{1}{\alpha}\psi'(u),$$

up to Lebesgue-null sets. Under the uniform law, both densities take the value $1/\alpha$ with probability α and 0 with probability $1-\alpha$. Thus, for either order,

$$D_f(\rho_o^\mu \| \lambda) = D_f(\rho_{o'}^\mu \| \lambda) = \alpha f(1/\alpha) + (1-\alpha) f(0) = D_f(\mu \| \nu).$$

In particular, both Kullback–Leibler divergences are $\log(1/\alpha)$, although one output density is concentrated on an upper tail and the other on many separated intervals.

Corollary 4.6 concerns the whole output, but it also holds for arbitrary finite prefixes $D \subseteq \mathcal{I}$ of the index order in o. For such D, write $\rho_{o,D}^\mu := (\pi_D)_\# \rho_o^\mu$ for the marginal of the output law on the coordinates in D.

Corollary 4.7 (Divergences along finite prefixes)**.** *For every finite prefix D,*

$$D_f\left(\rho_{o,D}^\mu \middle\| \lambda^D\right) = D_f(\mu_D \| \nu_D). \tag{10}$$

Corollary 4.7 provides an exact quantification of the claim in Corollary 2.8, guaranteeing not only that every departure from ν is visible in some finite prefix, but that the magnitude of departure matches that in the relevant indices of the original law.

As these finite prefixes grow to approximate the entire index set, the divergences also recover its full magnitude.

Corollary 4.8 (Recovery of the full divergence)**.** *Let $(D_m)_{m\geq 1}$ be an increasing sequence of finite prefixes with $\bigcup_m D_m = \mathcal{I}$. Then,*

$$D_f\left(\rho_{o,D_m}^\mu \middle\| \lambda^{D_m}\right) \uparrow D_f(\mu \| \nu) \qquad \text{in } [0,\infty]. \tag{11}$$

Example 4.9 (A single-coordinate mean shift under different orders)**.** Let $\mathcal{I} = \{1,\dots,d\}$ with $d \geq 2$, use the usual component orders on $\mathbb{R}$, and take

$$\nu = N(0, I_d), \qquad \mu = N(\delta e_j, I_d),$$

where e_j is the j-th standard basis vector and $\delta \neq 0$. Only coordinate j differs from the reference. Since ν is a product law, every index order gives the same transform up to null sets, $[T_o(x,v)]_k = \Phi(x_k)$, where Φ is the standard normal CDF.

For an order o, let D_m contain its first m indices and let $r_o(j)$ be the position of j in that order. For $j \notin D_m$ we have $\mu_{D_m} = \nu_{D_m}$, whereas once $j \in D_m$ the marginal likelihood ratio becomes $\exp(\delta x_j - \delta^2/2)$,

with μ-expected log-likelihood ratio $\delta^2/2$. Hence, Corollary 4.7 gives

$$D_{\mathrm{KL}}\left(\rho^{\mu}_{o,D_m}\middle\|\lambda^{D_m}\right) = D_{\mathrm{KL}}(\mu_{D_m}\|\nu_{D_m}) = \begin{cases} 0, & m < r_o(j), \\ \delta^2/2, & m \geq r_o(j). \end{cases}$$

If j is ordered first, the entire divergence is captured by the very first output coordinate, whereas if j is placed last, every proper prefix remains exactly uniform.

5 Discussion

The EGRT provides a common basis for goodness-of-fit testing on countable products of standard Borel spaces and, up to choice of order, is unique among a natural class of uniformizing transforms. The choice of order is an important design consideration. We show conditions under which the EGRT is invariant to this choice, but they are not guaranteed in practice.

Every order retains the same statistical information available to downstream testing, and we show how tests can be transferred between orders to produce equivalent diagnostics. However, transporting a test across orders can yield a considerably more intricate rejection rule or require access to additional coordinates.

In practice, the key question is therefore how to choose an order under which plausible departures from the reference model are readily detected by simple diagnostics. While one can mitigate sensitivity to any single ordering by using order-invariant test statistics (Dovern and Manner, 2020) or pooling evidence across multiple orders (Pournaderi, 2026), a systematic framework for selecting or combining orders to target specific alternatives remains an open problem.

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Supplementary material

S1 Notation

The table below collects the notation used in the main text and supplement.

Symbol	Meaning
Spaces, measures, and kernels	
$(\mathcal{X}_k, \mathcal{B}_k)$	Standard Borel component space at coordinate k.
$(\mathcal{X}_S, \mathcal{B}_S)$	Product space $\prod_{k\in S} \mathcal{X}_k$ with product σ-algebra, for $S \subseteq \mathcal{I}$; $(\mathcal{X}_\mathcal{I}, \mathcal{B}_\mathcal{I})$ denotes the full countable product.
x_S, X_S	Restriction of a point $x_\mathcal{I}$, or of the random element $X_\mathcal{I}$, to coordinates in S.
ν, μ	Reference and input probability measures on $\mathcal{X}_\mathcal{I}$, respectively.
λ, λ^S	Lebesgue measure on $[0,1]$ and its product on $[0,1]^S$; $\lambda^\mathcal{I}$ is the auxiliary-uniform law.
$Q_k^\nu(x_{\lhd k}, \cdot)$	Fixed version of the conditional law of X_k given its strict predecessors under the current finite-predecessor index order.
$Q_{k\mid S}^\nu(x_S, \cdot)$	Fixed version of the conditional law of X_k given X_S, for $S \subseteq \mathcal{I} \setminus \{k\}$.
$\mathcal{H}_k^\nu$	Family $\{Q_k^\nu(h, \cdot) : h \in \mathcal{X}_{\lhd k}\}$ for the fixed kernel version, including histories in $\nu_{\lhd k}$-null sets.
ν_S, μ_S	Marginals of ν and μ on coordinates $S \subseteq \mathcal{I}$.
$\mathcal{L}_\mu(Z)$	Law of Z when $X_\mathcal{I} \sim \mu$, with independent auxiliary uniforms understood.
Component-space orders	
$\preceq_k$, $\prec_k$	Borel-embeddable total order on $\mathcal{X}_k$ and its strict part.
$h_{\preceq_k}$	Borel order embedding of $(\mathcal{X}_k, \preceq_k)$ into $(\mathbb{R}, \leq)$.
$\preceq_{p,k}$	Partial order on $\mathcal{X}_k$.
$\vec{\preceq} = (\preceq_k)_{k\in\mathcal{I}}$	Sequence of component-space total orders.
$\mathcal{E}(\preceq_{p,k})$	Class of Borel-embeddable linear extensions of $\preceq_{p,k}$.
$I_{p,k}(x)$	Set of elements of $\mathcal{X}_k$ incomparable with x under $\preceq_{p,k}$.
$\Delta_k(x_{\trianglelefteq k})$	Conditional ambiguity $Q_k^\nu(x_{\lhd k}, I_{p,k}(x_k))$.

Symbol	Meaning
$(f_n)_{n\geq 1}$	Countable Borel monotone functions: point-separating in Theorem 3.2; representing the partial order through Equation (2) in Theorem 3.5.
Index-set orders and finite input blocks	
$\trianglelefteq$, $\triangleleft$	Total index order with finite strict predecessor sets, and its strict part.
$P_{\triangleleft}(k)$	Strict predecessor set $\{j \in \mathcal{I} : j \triangleleft k\}$.
$x_{\triangleleft k}$, $x_{\trianglelefteq k}$	Coordinates strictly preceding k, and those preceding or equal to k, under the current index order.
$\trianglelefteq_p$	Partial order on the index set $\mathcal{I}$.
$\mathcal{L}(\trianglelefteq_p)$	Finite-predecessor total-order linear extensions of $\trianglelefteq_p$.
$\mathrm{anc}_p(k)$	Strict ancestors of k under $\trianglelefteq_p$.
$\mathrm{inc}_p(k)$	Indices incomparable with k under $\trianglelefteq_p$.
$\mathrm{Pa}_G(k)$, $\mathrm{ND}_G(k)$	Parents and non-descendants of k in a DAG G on $\mathcal{I}$.
Transforms	
$F_\eta^{\preceq}(x)$, $F_\eta^{\preceq}(x^-)$	CDF and strict-left CDF of η under $\preceq$.
$\mathrm{rPIT}_\eta^{\preceq}(x, v)$	Randomized probability integral transform $F_\eta^{\preceq}(x^-) + v\eta(\{x\})$.
$T_{\mathcal{I},\nu}^{\vec{\preceq},\trianglelefteq}$	EGRT constructed from reference law ν, component orders $\vec{\preceq}$, and index order $\trianglelefteq$.
$U^{\vec{\preceq},\trianglelefteq}$	EGRT output $T_{\mathcal{I},\nu}^{\vec{\preceq},\trianglelefteq}(X_{\mathcal{I}}, V_{\mathcal{I}})$.
$T_{\mathcal{I},\nu}^{\vec{\preceq}}$, $U^{\vec{\preceq}}$	Notation when the index order is fixed and dependence on component orders is emphasized.
$T_{\mathcal{I},\nu}^{\trianglelefteq}$, $U^{\trianglelefteq}$	Notation when component orders are fixed and dependence on the index order is emphasized.
locally randomized triangular	The k-th output may depend on $X_{\trianglelefteq k}$ and the current V_k, but not on earlier auxiliary variables.
Reconstruction and statistical experiments	
$\mathfrak{O}$, T_o	Class of admissible order specifications and the EGRT under o.
P_o, ρ_o^μ	Forward kernel obtained by averaging over auxiliary uniforms, and output law $\mu\mathsf{P}_o$; $\mathsf{P}_o\psi$ is the conditional rejection probability of a transformed-space test ψ.
G_o, R_o	Measurable reconstruction map and its deterministic Markov kernel.

Symbol	Meaning
$\mathcal{M}$	Family of input laws dominated by ν.
R	Likelihood ratio $\mathrm{d}\mu/\mathrm{d}\nu$ for $\mu \ll \nu$.
$D_f(P\|Q)$	f-divergence $\int f(\mathrm{d}P/\mathrm{d}Q)\,\mathrm{d}Q$ for $P \ll Q$, with the generator specified in Section 4.2.
D, $\rho^{\mu}_{o,D}$	Finite downward-closed prefix and the corresponding EGRT output marginal $(\pi_D)_{\#}\rho^{\mu}_o$; $(D_m)_{m\geq 1}$ denotes an increasing sequence of such prefixes.

S2 Proofs for Section 2

Proof of Lemma 2.1.

Proof. Applying the standard disintegration theorem (Kallenberg, 2021, Theorem 8.5) to the pair of standard Borel random elements $(\pi_{\lhd k}, \pi_k)$ under ν yields the stated kernel and its $\nu_{\lhd k}$-a.e. uniqueness. When $P_{\lhd}(k)$ is empty, the conditioning space is a singleton and the kernel reduces to the marginal $\nu_{\{k\}}$. Because $(\pi_{\lhd k}, \pi_k)$ depends on the index order only through $P_{\lhd}(k)$, the same holds for the kernel up to $\nu_{\lhd k}$-null sets. □

Lemma S2.1 (Joint measurability of the conditional rPIT)**.** *Let $(S, \mathcal{S})$ be a measurable space, $(\mathcal{X}, \mathcal{B})$ a standard Borel space, $K : S \times \mathcal{B} \to [0,1]$ a probability kernel, and $\preceq$ a Borel-embeddable total order on $\mathcal{X}$ with Borel order embedding $h : \mathcal{X} \to \mathbb{R}$. Then the map*

$$(s, x, v) \longmapsto \mathrm{rPIT}^{\preceq}_{K(s,\cdot)}(x, v) = K(s, \{y \in \mathcal{X} : y \prec x\}) + vK(s, \{x\})$$

from $S \times \mathcal{X} \times [0,1]$ to $[0,1]$ is $\mathcal{S} \otimes \mathcal{B} \otimes \mathcal{B}([0,1])$-measurable.

Proof. Let

$$\mathcal{M} := \{A \in \mathcal{B} \otimes \mathcal{B} : (s, x) \mapsto K(s, A_x) \text{ is } \mathcal{S} \otimes \mathcal{B}\text{-measurable}\},$$

where $A_x := \{y \in \mathcal{X} : (x, y) \in A\}$. For any measurable rectangle $A = B_1 \times B_2$, we have $K(s, A_x) = \mathbf{1}_{B_1}(x)K(s, B_2)$, so $\mathcal{M}$ contains all measurable rectangles. Moreover, $\mathcal{M}$ is closed under complements since $(A^c)_x = (A_x)^c$ and $K(s, (A^c)_x) = 1 - K(s, A_x)$, and it is closed under countable disjoint unions because for disjoint $(A_n)_{n\geq 1} \subseteq \mathcal{M}$, $K(s, (\cup_n A_n)_x) = \sum_n K(s, (A_n)_x)$ is a pointwise limit of measurable partial sums. By the π–λ theorem, $\mathcal{M} = \mathcal{B} \otimes \mathcal{B}$.

Now consider the sets

$$L := \{(x, y) \in \mathcal{X}^2 : y \prec x\}, \qquad D := \{(x, y) \in \mathcal{X}^2 : y = x\}.$$

Because h is Borel, $L = (h \times h)^{-1}\{(a, b) \in \mathbb{R}^2 : b < a\}$ is Borel in $\mathcal{X}^2$, and the diagonal D is Borel since $\mathcal{X}$ is standard Borel. Applying the property defining $\mathcal{M}$ to L and D, we see that both

$$(s, x) \mapsto K(s, L_x) = K(s, \{y : y \prec x\}) \quad \text{and} \quad (s, x) \mapsto K(s, D_x) = K(s, \{x\})$$

are $\mathcal{S} \otimes \mathcal{B}$-measurable, from which joint measurability of $(s, x, v) \mapsto \mathrm{rPIT}^{\preceq}_{K(s,\cdot)}(x, v)$ is immediate. □

Lemma S2.2 (Measurable reconstruction from the conditional rPIT)**.** *Let $(S, \mathcal{S})$ and $(\mathcal{X}, \mathcal{B})$ be standard Borel spaces, $K : S \times \mathcal{B} \to [0, 1]$ a probability kernel, and $\preceq$ a Borel-embeddable total order on $\mathcal{X}$ with Borel order embedding $h : \mathcal{X} \to \mathbb{R}$. There exists a measurable map $G_K : S \times [0, 1] \to \mathcal{X}$ such that, for every probability measure ρ on S, if*

$$(S_0, X, V) \sim \rho(ds)K(s, dx)\lambda(dv),$$

then

$$G_K\big(S_0, \mathrm{rPIT}^{\preceq}_{K(S_0,\cdot)}(X, V)\big) = X \qquad a.s.$$

Moreover, for each $s \in S$, $(G_K(s, \cdot))_{\#}\lambda = K(s, \cdot)$.

Proof. By the Lusin–Souslin theorem (Kechris, 1995, Corollary 15.2), the image $Y := h(\mathcal{X})$ is Borel in $\mathbb{R}$ and $h^{-1} : Y \to \mathcal{X}$ is Borel. Let $K^h(s, \cdot) := h_{\#}K(s, \cdot)$ denote the pushforward kernel on $\mathbb{R}$, and define its cumulative distribution function $F_s(t) := K^h(s, (-\infty, t])$, which is measurable in s for each fixed t and right-continuous in t for each fixed s. Setting

$$H(s, u) := \begin{cases} \inf\{t \in \mathbb{Q} : F_s(t) \geq u\}, & 0 < u < 1, \\ 0, & u \in \{0, 1\}, \end{cases}$$

gives a jointly measurable generalized inverse on $S \times [0, 1]$ since the infimum is taken over a countable set. Fixing an arbitrary point $x_\star \in \mathcal{X}$, define

$$G_K(s, u) := \begin{cases} h^{-1}(H(s, u)), & H(s, u) \in Y, \\ x_\star, & H(s, u) \notin Y. \end{cases}$$

For each $s \in S$ and $U \sim \lambda$, we have $H(s,U) \sim K^h(s,\cdot)$. Because $K^h(s,Y) = 1$, the event $\{H(s,U) \notin Y\}$ is λ-null, and thus $(G_K(s,\cdot))_{\#}\lambda = K(s,\cdot)$ for all s.

To verify the almost-sure reconstruction property, note that under the embedding h,

$$\mathrm{rPIT}^{\preceq}_{K(s,\cdot)}(x,v) = F_s(h(x)^-) + vK^h(s,\{h(x)\}).$$

By the standard quantile identity for the real-valued randomized PIT (Rüschendorf, 2009, Proposition 2.1),

$$H\big(s, \mathrm{rPIT}^{\preceq}_{K(s,\cdot)}(X,V)\big) = h(X) \qquad \text{a.s.}$$

under $K(s,dx)\lambda(dv)$ for each $s \in S$. Applying h^{-1} and integrating with respect to $\rho(ds)$ yields the stated identity. Note that G_K is constructed purely from K and h, independently of ρ. □

Lemma S2.3 (Uniqueness of the rPIT under lexicographic monotonicity)**.** *Let $(\mathcal{X},\mathcal{B})$ be a standard Borel space with Borel-embeddable total order $\preceq$, and let η be a probability measure on $\mathcal{X}$. Suppose $S : \mathcal{X} \times [0,1] \to [0,1]$ is measurable, nondecreasing for the lexicographic order on $\mathcal{X} \times [0,1]$, and satisfies*

$$S_{\#}(\eta \otimes \lambda) = \lambda.$$

Then

$$S(x,v) = \mathrm{rPIT}^{\preceq}_{\eta}(x,v) \qquad (\eta \otimes \lambda)\text{-a.e.}$$

Proof. Via a Borel order embedding $h : \mathcal{X} \to \mathbb{R}$, we may push η and S forward to $h(\mathcal{X}) \subseteq \mathbb{R}$ and thus assume without loss of generality that $\mathcal{X} \subseteq \mathbb{R}$ with the standard order. Write $F(t) := \eta((-\infty,t])$, $F(t^-) := \eta((-\infty,t))$, and $\rho := \eta \otimes \lambda$.

Fix $a \in \mathbb{R}$ and put $L_a := (-\infty,a] \times [0,1]$, so that $\alpha_a := \rho(L_a) = F(a)$. Since every point in L_a lexicographically precedes every point in L_a^c, monotonicity of S implies that whenever $\rho(L_a) > 0$ and $\rho(L_a^c) > 0$,

$$s_a := \operatorname*{ess\,sup}_{L_a} S \le r_a := \operatorname*{ess\,inf}_{L_a^c} S.$$

We claim that

$$S \le \alpha_a \quad \rho\text{-a.e. on } L_a, \qquad S \ge \alpha_a \quad \rho\text{-a.e. on } L_a^c.$$

(If either L_a or L_a^c is ρ-null, the corresponding bound is vacuous and the other is trivial since $S \in [0,1]$.) Indeed, if $s_a > \alpha_a$, we can pick $t \in (\alpha_a, s_a)$ with $t \le r_a$; then $\{S \le t\} \subseteq L_a$ up to a ρ-null set, which gives $\rho\{S \le t\} \le \alpha_a < t$, contradicting $S_{\#}\rho = \lambda$. Symmetrically, if $r_a < \alpha_a$, picking $t \in (r_a, \alpha_a)$ with $t \ge s_a$ implies

that $L_a \subseteq \{S \le t\}$ up to a null set while $\rho(\{S \le t\} \cap L_a^c) > 0$, so $\rho\{S \le t\} > \alpha_a > t$, again a contradiction.

Intersecting these full-measure events over all rational $a \in \mathbb{Q}$, we obtain

$$F(x^-) \le S(x,v) \le F(x) \qquad \text{for } \rho\text{-a.e. } (x,v).$$

In particular, $S(x,v) = F(x) = F(x^-) = \mathrm{rPIT}_\eta(x,v)$ for ρ-a.e. (x,v) such that $\eta(\{x\}) = 0$.

It remains to handle the atoms of η, of which there are at most countably many. Fix an atom x and set $a := F(x^-)$, $m := \eta(\{x\}) > 0$, and $b := F(x) = a + m$. By the bounds above, $S(x,v) \in [a,b]$ for λ-a.e. v. For any $t \in [a,b)$, lexicographic monotonicity yields

$$\{S \le t\} = ((-\infty, x) \times [0,1]) \cup (\{x\} \times \{v : S(x,v) \le t\})$$

up to a ρ-null set, and therefore

$$t = \rho\{S \le t\} = a + m\lambda\{v : S(x,v) \le t\}.$$

That is, $v \mapsto S(x,v)$ pushes λ forward to $\mathrm{Unif}(a,b)$. Since $v \mapsto S(x,v)$ is nondecreasing, we must have $S(x,v) = a + mv = \mathrm{rPIT}_\eta(x,v)$ for λ-a.e. v. □

Proof of Theorem 2.4.

Proof. Let $h : \mathcal{X} \to \mathbb{R}$ be a Borel order embedding for $\preceq$, and write $\mu^h := h_\# \mu$ and $\nu^h := h_\# \nu$. Because

$$\{y \in \mathcal{X} : y \preceq x\} = h^{-1}((-\infty, h(x)]) \quad \text{and} \quad \{y \in \mathcal{X} : y \prec x\} = h^{-1}((-\infty, h(x))),$$

we have $\mathrm{rPIT}_\nu^{\preceq}(x,v) = \mathrm{rPIT}_{\nu^h}(h(x),v)$, where the right-hand side is the standard real-line randomized PIT. Under $(X,V) \sim \mu \otimes \lambda$, we have $(h(X),V) \sim \mu^h \otimes \lambda$, so the condition $(\mathrm{rPIT}_\nu^{\preceq})_\#(\mu \otimes \lambda) = \lambda$ is equivalent to

$$(\mathrm{rPIT}_{\nu^h})_\#(\mu^h \otimes \lambda) = \lambda.$$

Moreover, since h is a Borel isomorphism onto its image, $\mu = \nu$ if and only if $\mu^h = \nu^h$. When $\mu^h = \nu^h$, the displayed identity holds by standard uniformity of the real-valued randomized PIT (Rüschendorf, 2009, Proposition 2.1).

Conversely, suppose $(\mathrm{rPIT}_{\nu^h})_\#(\mu^h \otimes \lambda) = \lambda$, and let F_μ, F_ν denote the CDFs of μ^h, ν^h. Fix $t \in \mathbb{R}$, set

$u_0 := F_\nu(t)$, and define

$$A := (-\infty, t] \times [0,1], \qquad B := \{(y,v) : \mathrm{rPIT}_{\nu^h}(y,v) \le u_0\}.$$

Clearly $A \subseteq B$, since $y \le t$ implies $\mathrm{rPIT}_{\nu^h}(y,v) \le F_\nu(t) = u_0$. On the other hand, if $(y,v) \in B \setminus A$, then $y > t$ and hence $F_\nu(y^-) \ge u_0$, which forces $F_\nu(y^-) = u_0$ and $v\nu^h(\{y\}) = 0$. Among such points $y > t$, those with $\nu^h(\{y\}) > 0$ require $v = 0$ (a λ-null slice), whereas all $y > t$ with $\nu^h(\{y\}) = 0$ and $F_\nu(y^-) = u_0$ satisfy $\mathrm{rPIT}_{\nu^h}(y,v) = u_0$ for all v. Because the pushforward under $\mu^h \otimes \lambda$ is λ, which is atomless and thus assigns mass zero to $\{u_0\}$, μ^h must assign mass zero to $\{y > t : \nu^h(\{y\}) = 0,\, F_\nu(y^-) = u_0\}$. Consequently, $(\mu^h \otimes \lambda)(B \setminus A) = 0$, and thus

$$F_\mu(t) = (\mu^h \otimes \lambda)(A) = (\mu^h \otimes \lambda)(B) = \lambda([0, u_0]) = F_\nu(t)$$

for all $t \in \mathbb{R}$, proving $\mu^h = \nu^h$. □

For a finite $\trianglelefteq$-downward closed set $D \subseteq \mathcal{I}$, we write

$$T^{\vec{\preceq},\trianglelefteq}_{D,\nu_D} : \mathcal{X}_D \times [0,1]^D \to [0,1]^D$$

for the EGRT of ν_D under the restricted orders, defined using the fixed kernels Q^ν_k from the main text (which remain valid disintegration kernels for ν_D since $P_\trianglelefteq(k) \subseteq D$ for all $k \in D$). For $D \subseteq E \subseteq \mathcal{I}$, let $\pi^{\mathcal{X}}_D : \mathcal{X}_E \to \mathcal{X}_D$ and $\pi^{[0,1]}_D : [0,1]^E \to [0,1]^D$ denote the coordinate projections.

Proof of Theorem 2.6.

Proof. Following the sequential uniformization argument of Rüschendorf and de Valk (1993, Theorem 3), enumerate $\mathcal{I}$ in increasing $\trianglelefteq$-order as $k_1, \dots, k_n$ (or $k_1, k_2, \dots$ if $\mathcal{I}$ is infinite) and set $D_m := \{k_1, \dots, k_m\}$. We show by induction on m that

$$\big(T^{\vec{\preceq},\trianglelefteq}_{D_m,\nu_{D_m}}\big)_{\#}(\nu_{D_m} \otimes \lambda^{D_m}) = \lambda^{D_m}.$$

The base case $m = 1$ is Theorem 2.4 applied to $\nu_{\{k_1\}}$. For $m \ge 2$, disintegrate ν_{D_m} as

$$\nu_{D_m}(dx_{D_m}) = \nu_{D_{m-1}}(dx_{D_{m-1}}) Q^\nu_{k_m}(x_{D_{m-1}}, dx_{k_m}).$$

For any measurable rectangle $A \times B \subseteq [0,1]^{D_{m-1}} \times [0,1]$, Fubini's theorem, Theorem 2.4 applied conditionally

given $x_{D_{m-1}}$, and the induction hypothesis give

$$\begin{aligned}
&(\nu_{D_m} \otimes \lambda^{D_m}) \left\{ T^{\vec{\trianglelefteq},\trianglelefteq}_{D_m,\nu_{D_m}} \in A \times B \right\} \\
&\quad = \int \mathbf{1}_A \Big(T^{\vec{\trianglelefteq},\trianglelefteq}_{D_{m-1},\nu_{D_{m-1}}}(x_{D_{m-1}}, v_{D_{m-1}}) \Big) \, \lambda(B) \, \nu_{D_{m-1}}(dx_{D_{m-1}}) \lambda^{D_{m-1}}(dv_{D_{m-1}}) \\
&\quad = \lambda^{D_{m-1}}(A)\lambda(B).
\end{aligned}$$

By a monotone-class argument, the pushforward on D_m is λ^{D_m}.

This completes the proof when $\mathcal{I}$ is finite. When $\mathcal{I}$ is countably infinite, any finite subset $S \subseteq \mathcal{I}$ is contained in $D(S) := \bigcup_{k \in S} \{j \in \mathcal{I} : j \trianglelefteq k\}$, which is finite and downward closed by the finite-predecessor assumption. Uniformity on $D(S)$ implies uniformity on S by marginalization, and since S was arbitrary, the full pushforward law is $\lambda^{\mathcal{I}}$. □

Proposition S2.4 (Pointwise projective consistency with inherited kernels)**.** *Let $D \subseteq E \subseteq \mathcal{I}$ be finite $\trianglelefteq$-downward closed sets. Then, for every $(x_E, v_E) \in \mathcal{X}_E \times [0,1]^E$,*

$$\pi^{[0,1]}_D \Big(T^{\vec{\trianglelefteq},\trianglelefteq}_{E,\nu_E}(x_E, v_E) \Big) = T^{\vec{\trianglelefteq},\trianglelefteq}_{D,\nu_D}\big(\pi^{\mathcal{X}}_D x_E, \pi^{[0,1]}_D v_E\big).$$

Consequently, for every finite downward-closed $D \subseteq \mathcal{I}$ and every $(x_{\mathcal{I}}, v_{\mathcal{I}}) \in \mathcal{X}_{\mathcal{I}} \times [0,1]^{\mathcal{I}}$,

$$\pi^{[0,1]}_D \Big(T^{\vec{\trianglelefteq},\trianglelefteq}_{\mathcal{I},\nu}(x_{\mathcal{I}}, v_{\mathcal{I}}) \Big) = T^{\vec{\trianglelefteq},\trianglelefteq}_{D,\nu_D}\big(\pi^{\mathcal{X}}_D x_{\mathcal{I}}, \pi^{[0,1]}_D v_{\mathcal{I}}\big).$$

Proof of Proposition S2.4.

Proof. Because D is downward closed, $P_{\trianglelefteq}(k) \subseteq D$ for every $k \in D$, so k has the exact same predecessor set in D, E, and $\mathcal{I}$. Since all three transforms are defined using the same inherited kernel Q^{ν}_k and component order $\preceq_k$ at coordinate k, their k-th coordinates agree pointwise for every $k \in D$. □

Proof of Theorem 2.7.

Proof. The forward direction is Theorem 2.6. For the converse, fix a finite nonempty $\trianglelefteq$-downward closed set D, enumerated as $k_1 \triangleleft \cdots \triangleleft k_n$, and put $D_m := \{k_1, \ldots, k_m\}$. Assuming

$$\big(T^{\vec{\trianglelefteq},\trianglelefteq}_{D,\nu_D}\big)_{\#}(\mu_D \otimes \lambda^D) = \lambda^D,$$

we show that $\mu_{D_m} = \nu_{D_m}$ by induction on m. The case $m = 1$ follows directly from Theorem 2.4.

Suppose $m \geq 2$ and $\mu_{D_{m-1}} = \nu_{D_{m-1}}$. Let $R_m(x_{D_{m-1}}, \cdot)$ be a regular conditional distribution of X_{k_m} given $X_{D_{m-1}}$ under μ_{D_m}. Applying Lemma S2.2 recursively to the first $m-1$ coordinates yields a measurable map $S_{m-1} : [0,1]^{D_{m-1}} \to \mathcal{X}_{D_{m-1}}$ such that $(S_{m-1})_{\#}\lambda^{D_{m-1}} = \nu_{D_{m-1}}$ and $S_{m-1}(U_{D_{m-1}}) = X_{D_{m-1}}$ a.s. under $\nu_{D_{m-1}} \otimes \lambda^{D_{m-1}}$—and hence also under $\mu_{D_{m-1}} \otimes \lambda^{D_{m-1}}$, since $\mu_{D_{m-1}} = \nu_{D_{m-1}}$ by the induction hypothesis (with no domination assumption required).

Because $X_{k_m} \perp\!\!\!\perp V_{D_{m-1}} \mid X_{D_{m-1}}$, conditioning first on $(X_{D_{m-1}}, V_{D_{m-1}})$ shows that $R_m(S_{m-1}(u), \cdot)$ is a version of the conditional law of X_{k_m} given $U_{D_{m-1}} = u$. By hypothesis, $U_{k_m} = \mathrm{rPIT}^{\preceq_{k_m}}_{Q^{\nu}_{k_m}(X_{D_{m-1}}, \cdot)}(X_{k_m}, V_{k_m})$ is conditionally λ-distributed given $U_{D_{m-1}} = u$, so for $\lambda^{D_{m-1}}$-a.e. u,

$$\left(\mathrm{rPIT}^{\preceq_{k_m}}_{Q^{\nu}_{k_m}(S_{m-1}(u), \cdot)}\right)_{\#} (R_m(S_{m-1}(u), \cdot) \otimes \lambda) = \lambda.$$

By Theorem 2.4, $R_m(S_{m-1}(u), \cdot) = Q^{\nu}_{k_m}(S_{m-1}(u), \cdot)$ for $\lambda^{D_{m-1}}$-a.e. u. Pushing forward via S_{m-1}, we see that the two kernels agree for $\mu_{D_{m-1}}$-a.e. history, and thus $\mu_{D_m} = \nu_{D_m}$.

Finally, if $\left(T^{\vec{\preceq}, \trianglelefteq}_{\mathcal{I}, \nu}\right)_{\#}(\mu \otimes \lambda^{\mathcal{I}}) = \lambda^{\mathcal{I}}$, Proposition S2.4 implies that $\left(T^{\vec{\preceq}, \trianglelefteq}_{D, \nu_D}\right)_{\#}(\mu_D \otimes \lambda^D) = \lambda^D$ for every finite downward-closed $D \subseteq \mathcal{I}$, whence $\mu_D = \nu_D$ by the finite-prefix result. Since any finite $S \subseteq \mathcal{I}$ is contained in the finite downward-closed set $D(S) := \bigcup_{k \in S} \{j : j \trianglelefteq k\}$, we have $\mu_S = \nu_S$ for all finite $S \subseteq \mathcal{I}$, and therefore $\mu = \nu$. □

Proof of Corollary 2.8.

Proof. Let $\rho := \left(T^{\vec{\preceq}, \trianglelefteq}_{\mathcal{I}, \nu}\right)_{\#}(\mu \otimes \lambda^{\mathcal{I}})$. Since $\mu \neq \nu$, Theorem 2.7 implies $\rho \neq \lambda^{\mathcal{I}}$, so $(\pi_S)_{\#}\rho \neq \lambda^S$ for some finite $S \subseteq \mathcal{I}$. Taking $D(S) := \bigcup_{k \in S} \{j : j \trianglelefteq k\}$ gives a finite downward-closed set on which $(\pi_{D(S)})_{\#}\rho \neq \lambda^{D(S)}$. □

Proof of Theorem 2.11.

Proof. By construction and Theorem 2.6, the EGRT $G := T^{\vec{\preceq}, \trianglelefteq}_{\mathcal{I}, \nu}$ satisfies properties (i)–(iii). Conversely, let $\widetilde{T} : \mathcal{X}_{\mathcal{I}} \times [0,1]^{\mathcal{I}} \to [0,1]^{\mathcal{I}}$ satisfy (i)–(iii). Enumerating $\mathcal{I}$ in increasing $\trianglelefteq$-order as $k_1, k_2, \ldots$ (or $k_1, \ldots, k_n$) and setting $D_m := \{k_1, \ldots, k_m\}$, we prove by induction on m that $\widetilde{T}_{D_m} = G_{D_m}$ $\nu \otimes \lambda^{\mathcal{I}}$-a.e.

For $m = 1$, triangularity and local randomization imply that $\widetilde{T}_{k_1}$ is a function of (X_{k_1}, V_{k_1}) alone; it is lexicographically nondecreasing and pushes $\nu_{\{k_1\}} \otimes \lambda$ forward to λ. By Lemma S2.3, $\widetilde{T}_{k_1}(X, V) = \mathrm{rPIT}^{\preceq_{k_1}}_{\nu_{\{k_1\}}}(X_{k_1}, V_{k_1}) = G_{k_1}(X, V)$ a.s.

Now assume $\widetilde{T}_{D_{m-1}} = G_{D_{m-1}}$ a.s., and write $k := k_m$. By Lemma S2.2, there is a measurable reconstruction map $S_{m-1} : [0,1]^{D_{m-1}} \to \mathcal{X}_{D_{m-1}}$ with $(S_{m-1})_{\#}\lambda^{D_{m-1}} = \nu_{D_{m-1}}$ and $S_{m-1}(G_{D_{m-1}}(X, V)) = X_{D_{m-1}}$ a.s. Because $\widetilde{T}$ has product-uniform output and $\widetilde{T}_{D_{m-1}} = G_{D_{m-1}}$ a.s., the conditional law of $\widetilde{T}_k$ given $\widetilde{T}_{D_{m-1}} = u$ is λ for $\lambda^{D_{m-1}}$-a.e. u, and $X_{D_{m-1}} = S_{m-1}(u)$ a.s. under this conditioning. Moreover, since $X_k \perp\!\!\!\perp V_{D_{m-1}} \mid X_{D_{m-1}}$

and $V_k \perp\!\!\!\perp (X_{\mathcal{I}}, V_{D_{m-1}})$, the conditional law of (X_k, V_k) given $\widetilde{T}_{D_{m-1}} = u$ is $Q_k^\nu(S_{m-1}(u), dx_k)\lambda(dv_k)$, while local randomization ensures $\widetilde{T}_k$ does not depend on $V_{D_{m-1}}$. For $\lambda^{D_{m-1}}$-a.e. u, the map $(x_k, v_k) \mapsto \widetilde{T}_k(S_{m-1}(u), x_k, v_k)$ is lexicographically nondecreasing and pushes $Q_k^\nu(S_{m-1}(u), \cdot) \otimes \lambda$ forward to λ, so Lemma S2.3 gives

$$\widetilde{T}_k(S_{m-1}(u), x_k, v_k) = \mathrm{rPIT}^{\preceq_k}_{Q_k^\nu(S_{m-1}(u),\cdot)}(x_k, v_k)$$

for $Q_k^\nu(S_{m-1}(u), dx_k)\lambda(dv_k)$-a.e. (x_k, v_k). Integrating over $u \sim \lambda^{D_{m-1}}$ yields $\widetilde{T}_k(X, V) = G_k(X, V)$ a.s., completing the induction. Taking the union of the null exceptional sets over m gives $\widetilde{T} = G$ $\nu \otimes \lambda^{\mathcal{I}}$-a.e. □

S3 Proofs for Section 3

Proof of Proposition 3.1.

Proof. Let $\preceq$ be a linear extension of $\preceq_p$, and enumerate $\mathcal{X} = \{x_1, x_2, \ldots\}$ (with the obvious modification if $\mathcal{X}$ is finite). We define an order embedding $\phi : (\mathcal{X}, \preceq) \to (\mathbb{Q}, \leq)$ inductively. Having defined ϕ on $\{x_1, \ldots, x_n\}$, set

$$L_n := \{\phi(x_i) : i \leq n,\ x_i \prec x_{n+1}\}, \qquad U_n := \{\phi(x_i) : i \leq n,\ x_{n+1} \prec x_i\}.$$

By the induction hypothesis, $\max L_n < \min U_n$ whenever both sets are nonempty; we may therefore pick $\phi(x_{n+1}) \in \mathbb{Q}$ strictly between $\max L_n$ and $\min U_n$ (treating an empty set as imposing no bound). By construction, $x \preceq y \iff \phi(x) \leq \phi(y)$, and since every map on a countable standard Borel space is Borel, ϕ is a Borel order embedding. □

Proof of Theorem 3.2. The geometric-series construction below adapts Hack et al. (2022, Lemmas 1–3) to the Borel setting.

Proof. (iii)⇒(ii): Let $(f_n)_{n\geq 1}$ be a countable point-separating family of monotone Borel functions. Enumerate $\mathbb{N} \times \mathbb{Q}$ as $((n_j, r_j))_{j\geq 1}$ and define

$$h(x) := \sum_{j=1}^{\infty} 3^{-j} \mathbf{1}\{f_{n_j}(x) > r_j\}.$$

Because the series converges uniformly, h is Borel. If $x \neq y$, point separation gives $f_n(x) \neq f_n(y)$ for some n, and choosing a rational r between $f_n(x)$ and $f_n(y)$ ensures the two series differ in at least one term. At the first index j_0 where they differ, the leading gap 3^{-j_0} exceeds the maximum possible tail $\sum_{j>j_0} 3^{-j} = 3^{-j_0}/2$, so $h(x) \neq h(y)$. Monotonicity of each f_n implies that if $x \preceq_p y$, then $\mathbf{1}\{f_{n_j}(x) > r_j\} \leq \mathbf{1}\{f_{n_j}(y) > r_j\}$ for all j, giving $h(x) \leq h(y)$; if $x \prec_p y$, then $h(x) < h(y)$ by injectivity. (Conversely, if $h : \mathcal{X} \to \mathbb{R}$ satisfies (ii), the singleton family $\{h\}$ immediately gives (iii).)

(ii)⇒(i): Given h as in (ii), define $x \preceq_h y \iff h(x) \le h(y)$. Since h is injective, $\preceq_h$ is a total order extending $\preceq_p$ with Borel order embedding h, which also establishes the final claim of the theorem.

(i)⇒(ii): If $\preceq \in \mathcal{E}(\preceq_p)$ has Borel order embedding $h_\preceq$, then $x \prec_p y$ implies $x \prec y$ and thus $h_\preceq(x) < h_\preceq(y)$. □

Lemma S3.1 (Measurability of conditional ambiguity)**.** *In the setting of Definition 3.7, the map*

$$\Delta_k(x_{\trianglelefteq k}) = Q_k^\nu(x_{\triangleleft k}, I_{p,k}(x_k))$$

is $\mathcal{B}_{\trianglelefteq k}$-measurable.

Proof. Since the graph $G_k := \{(x, y) : x \preceq_{p,k} y\}$ is Borel by assumption, the incomparability relation $J_k := (\mathcal{X}_k \times \mathcal{X}_k) \setminus (G_k \cup G_k^\top)$ is also Borel, with x-section $I_{p,k}(x)$. Viewing $Q_k^\nu(x_{\triangleleft k}, \mathrm{d}y)$ as a kernel $\widetilde{Q}_k^\nu((x_{\triangleleft k}, x_k), \mathrm{d}y)$ on $\mathcal{X}_{\trianglelefteq k} \times \mathcal{B}_k$, standard measurability of kernel integrals shows that

$$(x_{\triangleleft k}, x_k) \longmapsto \int \mathbf{1}_{J_k}(x_k, y)\, Q_k^\nu(x_{\triangleleft k}, \mathrm{d}y) = \Delta_k(x_{\trianglelefteq k})$$

is $\mathcal{B}_{\trianglelefteq k}$-measurable. □

Lemma S3.2 (CDF separation on countable spaces)**.** *Let $(\mathcal{X}, \mathcal{B})$ be a countable standard Borel space with partial order $\preceq_p$. Then $\mathcal{E}(\preceq_p)$ is CDF-separating. A countable separating family can be chosen to contain, for every ordered incomparable pair (a, b), an extension with $a \prec b$. If $\mathcal{X}$ is finite, the finite family of all linear extensions may be used.*

Proof. For each ordered incomparable pair (a, b), adjoining $a \prec b$ and taking the transitive closure introduces no cycles, so Szpilrajn's theorem yields a linear extension $\preceq_{a,b}$, which is Borel-embeddable by Proposition 3.1. Because $\mathcal{X}$ is countable, the collection of such extensions $\preceq_{a,b}$ (augmented by a single extension if $\preceq_p$ is already total) is countable.

If $\eta\{x : \eta(I_p(x)) > 0\} > 0$, countability of $\mathcal{X}$ ensures there exist incomparable $x, y \in \mathcal{X}$ with $\eta(\{x\}) > 0$ and $\eta(\{y\}) > 0$. Setting $\preceq^+ := \preceq_{x,y}$ and $\preceq^- := \preceq_{y,x}$, we have

$$F_\eta^{\preceq^+}(y) - F_\eta^{\preceq^+}(x) \ge \eta(\{y\}) > 0, \qquad F_\eta^{\preceq^-}(y) - F_\eta^{\preceq^-}(x) \le -\eta(\{x\}) < 0,$$

so $F_\eta^{\preceq^+}$ and $F_\eta^{\preceq^-}$ cannot agree at both x and y. Because the point masses $\eta(\{x\})$ and $\eta(\{y\})$ do not depend on the order, the strict-left CDFs likewise differ on a set of positive η-measure. When $\mathcal{X}$ is finite, the set of all linear extensions is finite and contains $\preceq^+$ and $\preceq^-$. □

Proof of Theorem 3.5. While Hack et al. (2022, Proposition 6) relate countable multi-utilities to countable families of injective monotones, we must show here that Borel extensions separate CDFs in the presence of positive incomparability mass (even for atomless laws) and that this probabilistic separation recovers the order representation.

Proof. (i)⇒(ii): By Equation (2), the graph of $\preceq_p$ is $\bigcap_{n\geq 1}\{(x,y) : f_n(x) \leq f_n(y)\}$ for a sequence of $\preceq_p$-monotone Borel functions $(f_n)_{n\geq 1}$. By antisymmetry of $\preceq_p$, $(f_n)_{n\geq 1}$ separates points, so Theorem 3.2 (composed with a strictly increasing bijection $\mathbb{R} \to (0,1)$) yields a Borel injection $h_0 : \mathcal{X} \to (0,1)$ satisfying $x \prec_p y \Rightarrow h_0(x) < h_0(y)$.

For each $n \geq 1$ and $q \in \mathbb{Q}$, define $h_{n,q} := \mathbf{1}\{f_n > q\} + h_0$. Since $h_0(\mathcal{X}) \subseteq (0,1)$, $h_{n,q}$ maps $\{f_n \leq q\}$ into $(0,1)$ and $\{f_n > q\}$ into $(1,2)$, making $h_{n,q}$ Borel and injective; furthermore, $x \prec_p y$ implies $h_{n,q}(x) < h_{n,q}(y)$. Thus its pullback order $\preceq_{n,q}$ belongs to $\mathcal{E}(\preceq_p)$, and $\mathcal{E}_0 := \{\preceq_{n,q} : n \geq 1,\ q \in \mathbb{Q}\}$ is a countable subfamily of $\mathcal{E}(\preceq_p)$.

If $x, y \in \mathcal{X}$ are incomparable, Equation (2) gives n, m with $f_n(x) > f_n(y)$ and $f_m(y) > f_m(x)$. Picking rationals q, q' strictly between these pairs of values, we see that $\preceq_{n,q}$ and $\preceq_{m,q'}$ order x and y in opposite directions. Hence the incomparability set $J := \{(x,y) : y \in I_p(x)\}$ is covered by the countable union

$$J \subseteq \bigcup_{\preceq,\preceq' \in \mathcal{E}_0} R_{\preceq,\preceq'}, \qquad R_{\preceq,\preceq'} := \{(x,y) : x \prec y,\ y \prec' x\}.$$

Now let η be any probability measure with $\eta\{x : \eta(I_p(x)) > 0\} > 0$. By Fubini's theorem, $(\eta \otimes \eta)(J) > 0$, so $(\eta \otimes \eta)(R_{\preceq,\preceq'}) > 0$ for some pair $\preceq, \preceq' \in \mathcal{E}_0$. Suppose for contradiction that $F_\eta^{\preceq}(\cdot^-) = F_\eta^{\preceq'}(\cdot^-)$ η-a.e.; since singleton masses are order-independent, we then have $\mathrm{rPIT}_\eta^{\preceq}(x,v) = \mathrm{rPIT}_\eta^{\preceq'}(x,v)$ for all $v \in [0,1]$ and η-a.e. x. Let $(X,Y,V,W) \sim \eta \otimes \eta \otimes \lambda \otimes \lambda$, and put

$$U := \mathrm{rPIT}_\eta^{\preceq}(X,V), \qquad Z := \mathrm{rPIT}_\eta^{\preceq}(Y,W).$$

By Theorem 2.4, U and Z are independent $\mathrm{Unif}(0,1)$ random variables. Yet whenever $x \prec y$,

$$\mathrm{rPIT}_\eta^{\preceq}(x,v) \leq F_\eta^{\preceq}(x) \leq F_\eta^{\preceq}(y^-) \leq \mathrm{rPIT}_\eta^{\preceq}(y,w) \qquad \text{for all } v,w \in [0,1],$$

which implies $U \leq Z$ on $\{(X,Y) \in R_{\preceq,\preceq'}\}$. Applying the same inequality to $\preceq'$ (where $Y \prec' X$) and using the a.s. agreement of the two rPITs gives $Z \leq U$ a.s. on $\{(X,Y) \in R_{\preceq,\preceq'}\}$. Thus $U = Z$ on an event of positive probability, which is impossible for independent continuous random variables. Hence $\mathcal{E}_0$ witnesses CDF separation for every probability measure η.

(ii)⇒(i): Let $\mathcal{E}_0 \subseteq \mathcal{E}(\preceq_p)$ be a nonempty countable family witnessing CDF separation for all probability laws, and pick a Borel order embedding $f_\preceq : \mathcal{X} \to \mathbb{R}$ for each $\preceq \in \mathcal{E}_0$. For any incomparable pair x, y, apply CDF separation to $\eta = (\delta_x + \delta_y)/2$. Since $\eta(I_p(x)) = \eta(I_p(y)) = 1/2 > 0$, the strict-left CDFs cannot take identical values on $\{x, y\}$ across all $\preceq \in \mathcal{E}_0$, meaning $\mathcal{E}_0$ must contain extensions ordering (x, y) in both directions; in particular, $f_\preceq(x) > f_\preceq(y)$ for some $\preceq \in \mathcal{E}_0$ and $f_{\preceq'}(x) < f_{\preceq'}(y)$ for another. Consequently,

$$x \preceq_p y \quad \iff \quad f_\preceq(x) \le f_\preceq(y) \text{ for every } \preceq \in \mathcal{E}_0,$$

and enumerating $\{f_\preceq : \preceq \in \mathcal{E}_0\}$ gives (i). □

Proof of Proposition 3.8.

Proof. Fix $k \in \mathcal{I}$, two extensions $\preceq, \preceq'$ of $\preceq_{p,k}$, a probability measure η on $\mathcal{X}_k$, and $x \in \mathcal{X}_k$. Setting $L := \{y : y \prec x\}$ and $L' := \{y : y \prec' x\}$, note that any $y \in L \triangle L'$ is ordered oppositely relative to x by $\preceq$ and $\preceq'$, so $L \triangle L' \subseteq I_{p,k}(x)$. Because $\eta(\{x\})$ does not depend on the order, we have for all $v \in [0, 1]$,

$$\left|\mathrm{rPIT}_\eta^{\preceq}(x, v) - \mathrm{rPIT}_\eta^{\preceq'}(x, v)\right| = |\eta(L) - \eta(L')| \le \eta(L \triangle L') \le \eta\big(I_{p,k}(x)\big). \tag{12}$$

Taking $\eta = Q_k^\nu(x_{\lhd k}, \cdot)$ and $x = x_k$ in Equation (12) yields the pointwise bound in the proposition. When $\Delta_k = 0$ $\nu_{\trianglelefteq k}$-a.e. for all $k \in \mathcal{I}$, the two transforms agree coordinatewise $\nu \otimes \lambda^{\mathcal{I}}$-a.e., and hence globally $\nu \otimes \lambda^{\mathcal{I}}$-a.e. since $\mathcal{I}$ is countable; by absolute continuity, this a.e. equality and equality of the pushforward laws carry over to every $\mu \ll \nu$. □

Proof of Theorem 3.10.

Proof. The chain (i)⇒(ii)⇒(iii) follows directly from Proposition 3.8.

To prove (iii)⇒(i), suppose by contraposition that (i) fails at coordinate k. Writing $h = x_{\lhd k}$ and $\eta_h = Q_k^\nu(h, \cdot)$, we have

$$\int \eta_h\{x : \eta_h(I_{p,k}(x)) > 0\}\, \nu_{\lhd k}(\mathrm{d}h) > 0,$$

so $\{h : \eta_h\{x : \eta_h(I_{p,k}(x)) > 0\} > 0\}$ has positive $\nu_{\lhd k}$-measure. Let $\mathcal{E}_0(\preceq_{p,k})$ be a countable subfamily witnessing CDF separation for $\mathcal{H}_k^\nu$ in Equation (3). Because $\mathcal{E}_0(\preceq_{p,k})$ is countable, there is a fixed pair $\preceq_k, \preceq_k' \in \mathcal{E}_0(\preceq_{p,k})$ whose strict-left CDFs under η_h differ on a set of positive η_h-measure for a set of histories h of positive $\nu_{\lhd k}$-measure. By Lemma S2.1 and Fubini's theorem, the set

$$D := \{(h, x_k) : F_{\eta_h}^{\preceq_k}(x_k^-) \ne F_{\eta_h}^{\preceq_k'}(x_k^-)\}$$

is measurable with $\nu_{\trianglelefteq k}(D) > 0$. Extend $\preceq_k$ and $\preceq'_k$ to full component-order sequences $\vec{\preceq}$ and $\vec{\preceq}'$ that coincide at all coordinates other than k.

For $z = (h, x_k)$ and Borel $B \subseteq [0,1]$, define

$$K(z,B) := \lambda\{v : T_k^{\vec{\preceq}}(z,v) \in B\}, \qquad K'(z,B) := \lambda\{v : T_k^{\vec{\preceq}'}(z,v) \in B\}.$$

For every $z \in D$, $K(z,\cdot)$ and $K'(z,\cdot)$ are uniform distributions (or point masses if $\eta_h(\{x_k\}) = 0$) on intervals of equal length $\eta_h(\{x_k\})$ with distinct left endpoints, so $K(z,\cdot) \neq K'(z,\cdot)$. Taking a countable algebra $\mathcal{A}_0$ that generates $\mathcal{B}([0,1])$, we have $D \subseteq \bigcup_{B\in\mathcal{A}_0}\{z : K(z,B) \neq K'(z,B)\}$. Swapping $\preceq_k$ and $\preceq'_k$ if necessary, we can find $B \in \mathcal{A}_0$ such that $D_B := \{z : K(z,B) > K'(z,B)\}$ has $\nu_{\trianglelefteq k}(D_B) > 0$. Setting $A := \{x_{\mathcal{I}} \in \mathcal{X}_{\mathcal{I}} : x_{\trianglelefteq k} \in D_B\}$ gives $\nu(A) = \nu_{\trianglelefteq k}(D_B) > 0$, and under $\nu(\cdot \mid A)$ the k-th coordinate marginals of the two transforms assign to B the strictly unequal probabilities

$$\frac{1}{\nu_{\trianglelefteq k}(D_B)} \int_{D_B} K(z,B)\, \nu_{\trianglelefteq k}(\mathrm{d}z) > \frac{1}{\nu_{\trianglelefteq k}(D_B)} \int_{D_B} K'(z,B)\, \nu_{\trianglelefteq k}(\mathrm{d}z).$$

The joint output laws therefore differ under $\mu = \nu(\cdot \mid A) \ll \nu$, contradicting (iii). □

Lemma S3.3 (Existence of finite-predecessor linear extensions)**.** *Let $\trianglelefteq_p$ be a partial order on a countable set $\mathcal{I}$ such that every* $\mathrm{anc}_p(k)$ *is finite. Then $\mathcal{L}(\trianglelefteq_p)$ is nonempty.*

Proof. Fix an enumeration $\mathcal{I} = \{i_1, i_2, \ldots\}$ and recursively construct a sequence by selecting at each step the $\trianglelefteq_p$-minimal remaining element of $\mathcal{I}$ with the smallest enumeration index. (A minimal remaining element always exists, as an infinite descending chain would yield infinitely many ancestors.) By induction on $|\mathrm{anc}_p(k)|$, every $k \in \mathcal{I}$ is eventually selected: once all elements of $\mathrm{anc}_p(k)$ have been chosen, k becomes minimal among the remaining elements and can be preceded by at most finitely many indices with smaller enumeration index. Ordering $\mathcal{I}$ by selection time gives an extension in $\mathcal{L}(\trianglelefteq_p)$. □

Lemma S3.4 (Control of predecessor sets in index extensions)**.** *Under the assumptions of Lemma S3.3, fix $k \in \mathcal{I}$.*

(a) There exists $\trianglelefteq_k^0 \in \mathcal{L}(\trianglelefteq_p)$ such that

$$P_{\trianglelefteq_k^0}(k) = \mathrm{anc}_p(k).$$

(b) For every finite $B \subseteq \mathrm{inc}_p(k)$, there exists $\trianglelefteq^{k,B} \in \mathcal{L}(\trianglelefteq_p)$ such that

$$B \subseteq P_{\trianglelefteq^{k,B}}(k).$$

Proof. (a) Adjoin to $\trianglelefteq_p$ the relations $k \triangleleft j$ for all $j \in \mathrm{inc}_p(k)$ and take the transitive closure. Because every added relation exits k and no $j \in \mathrm{inc}_p(k)$ is an ancestor of k, no cycle can arise. Under this enlarged partial order, the ancestor set of any index is contained in its original ancestor set united with $\mathrm{anc}_p(k) \cup \{k\}$, which is finite. Applying Lemma S3.3 to the enlarged partial order yields $\trianglelefteq_k^0 \in \mathcal{L}(\trianglelefteq_p)$ in which every $j \notin \mathrm{anc}_p(k) \cup \{k\}$ follows k, so $P_{\triangleleft_k^0}(k) = \mathrm{anc}_p(k)$.

(b) Similarly, adjoining $b \triangleleft k$ for each $b \in B$ and taking the transitive closure creates no cycles, and increases any ancestor set by at most the finite set $\bigcup_{b \in B}(\mathrm{anc}_p(b) \cup \{b\})$. Any extension in $\mathcal{L}$ of this enlarged partial order (via Lemma S3.3) belongs to $\mathcal{L}(\trianglelefteq_p)$ and satisfies $B \subseteq P_{\triangleleft^{k,B}}(k)$. □

Proof of Theorem 3.11.

Proof. Fix $k \in \mathcal{I}$ and put $A := \mathrm{anc}_p(k)$ and $C := \mathrm{inc}_p(k)$. For any $\trianglelefteq \in \mathcal{L}(\trianglelefteq_p)$, $S := P_{\triangleleft}(k)$ satisfies $A \subseteq S \subseteq A \cup C$.

(i)⇔(ii): If (i) holds, decomposition of conditional independence yields $X_k \perp\!\!\!\perp_\nu X_{S \setminus A} \mid X_A$, which is equivalent to the kernel identity in (ii). Conversely, assume (ii) and let $B \subseteq C$ be finite. By Lemma S3.4(b), there exists $\trianglelefteq^{k,B} \in \mathcal{L}(\trianglelefteq_p)$ with $B \subseteq P_{\triangleleft^{k,B}}(k) \setminus A$, so (ii) gives $X_k \perp\!\!\!\perp_\nu X_{P_{\triangleleft^{k,B}}(k) \setminus A} \mid X_A$ and hence $X_k \perp\!\!\!\perp_\nu X_B \mid X_A$. Since $B \subseteq C$ was an arbitrary finite subset and C is countable, a monotone-class argument yields $X_k \perp\!\!\!\perp_\nu X_C \mid X_A$.

(ii)⇔(iii): Under (ii), for every $\trianglelefteq \in \mathcal{L}(\trianglelefteq_p)$ and $k \in \mathcal{I}$,

$$[T^{\trianglelefteq}_{\mathcal{I},\nu}(X,V)]_k = \mathrm{rPIT}^{\trianglelefteq_k}_{Q^\nu_{k|A}(X_A,\cdot)}(X_k, V_k) \qquad \nu \otimes \lambda^{\mathcal{I}}\text{-a.e.}$$

Since the right-hand side is independent of $\trianglelefteq$ and $\mathcal{I}$ is countable, (iii) follows. Conversely, suppose (iii) holds and fix $\trianglelefteq \in \mathcal{L}(\trianglelefteq_p)$ with $S = P_{\triangleleft}(k)$. Comparing $\trianglelefteq$ against the extension $\trianglelefteq_k^0$ from Lemma S3.4(a) (for which $P_{\triangleleft_k^0}(k) = A$) gives

$$\begin{aligned} &\mathrm{rPIT}^{\trianglelefteq_k}_{Q^\nu_{k|S}(X_S,\cdot)}(X_k, V_k) \\ &\qquad = \mathrm{rPIT}^{\trianglelefteq_k}_{Q^\nu_{k|A}(X_A,\cdot)}(X_k, V_k) \qquad \nu \otimes \lambda^{\mathcal{I}}\text{-a.e.} \end{aligned}$$

Disintegrating over $X_S = s$, we see that for ν_S-a.e. s, the two conditional rPITs agree $Q^\nu_{k|S}(s, \mathrm{d}x)\lambda(\mathrm{d}v)$-a.e. Because the left-hand rPIT pushes $Q^\nu_{k|S}(s,\cdot) \otimes \lambda$ forward to λ, so does the right-hand rPIT, and Theorem 2.4 therefore implies $Q^\nu_{k|S}(s,\cdot) = Q^\nu_{k|A}(s_A,\cdot)$ for ν_S-a.e. s, which is (ii).

(iii)⇒(iv): Absolute continuity $\mu \ll \nu$ transfers $\nu \otimes \lambda^{\mathcal{I}}$-a.e. map equality to $\mu \otimes \lambda^{\mathcal{I}}$, giving (iv) as well as the final statement of the theorem.

(iv)⇒(iii): Suppose (iii) fails for some $\trianglelefteq, \trianglelefteq' \in \mathcal{L}(\trianglelefteq_p)$. Since $\mathcal{I}$ is countable, the two transforms differ at some coordinate k on a set of positive $\nu \otimes \lambda^{\mathcal{I}}$-measure. Set $S := P_{\trianglelefteq}(k)$, $S' := P_{\trianglelefteq'}(k)$, and $H := S \cup S' \cup \{k\}$. For $z \in \mathcal{X}_H$ and $v \in [0,1]$, define the affine fiber maps

$$\phi_z(v) := \mathrm{rPIT}^{\prec_k}_{Q^{\nu}_{k|S}(z_S,\cdot)}(z_k, v), \qquad \phi'_z(v) := \mathrm{rPIT}^{\prec_k}_{Q^{\nu}_{k|S'}(z_{S'},\cdot)}(z_k, v),$$

with pushforward laws $K(z,\cdot) := (\phi_z)_{\#}\lambda$ and $K'(z,\cdot) := (\phi'_z)_{\#}\lambda$. Because $v \mapsto \phi_z(v)$ is nondecreasing and affine of the form $a + mv$ ($m \geq 0$), $K(z,\cdot)$ is the uniform law on $[a, a+m]$ (or δ_a if $m = 0$), which uniquely determines a and m. By Fubini's theorem, $D_0 := \{z \in \mathcal{X}_H : \lambda\{v : \phi_z(v) \neq \phi'_z(v)\} > 0\}$ has $\nu_H(D_0) > 0$, and $K(z,\cdot) \neq K'(z,\cdot)$ for all $z \in D_0$. Letting $\mathcal{A}_0$ be a countable algebra generating $\mathcal{B}([0,1])$, we can find $B \in \mathcal{A}_0$ (interchanging $\trianglelefteq$ and $\trianglelefteq'$ if needed) such that $D_B := \{z \in \mathcal{X}_H : K(z,B) > K'(z,B)\}$ satisfies $\nu_H(D_B) > 0$. Then $E := \{x_{\mathcal{I}} \in \mathcal{X}_{\mathcal{I}} : x_H \in D_B\}$ has $\nu(E) = \nu_H(D_B) > 0$, and under $\nu(\cdot \mid E)$ the k-th output marginals of the two transforms assign unequal probabilities

$$\frac{1}{\nu_H(D_B)} \int_{D_B} K(z,B)\, \nu_H(\mathrm{d}z) > \frac{1}{\nu_H(D_B)} \int_{D_B} K'(z,B)\, \nu_H(\mathrm{d}z)$$

to B. Hence the joint output laws differ under $\mu = \nu(\cdot \mid E) \ll \nu$, contradicting (iv). □

Proof of Corollary 3.12.

Proof. Fix $k \in \mathcal{I}$ and put $A := \mathrm{anc}_p(k)$, $C := \mathrm{inc}_p(k)$, and $P := \mathrm{Pa}_G(k)$, so that $P \subseteq A$ and $\mathrm{ND}_G(k) = A \cup C$ (disjoint). By the local Markov property,

$$Q^{\nu}_{k|A\cup C}(X_{A\cup C}, \cdot) = Q^{\nu}_{k|P}(X_P, \cdot) \qquad \nu\text{-a.s.}$$

Taking conditional expectations given X_A and using $P \subseteq A$, we also have $Q^{\nu}_{k|A}(X_A,\cdot) = Q^{\nu}_{k|P}(X_P,\cdot)$ ν-a.s., whence $X_k \perp\!\!\!\perp_{\nu} X_C \mid X_A$. Since every $k \in \mathcal{I}$ has finite ancestor set by assumption, Theorem 3.11 applies. □

S4 Proofs for Section 4

Proof of Lemma 4.1.

Proof. Following the recursive quantile-inverse construction of Rüschendorf and de Valk (1993, Theorem 3) (see also Rüschendorf, 2009, Section 3), enumerate $\mathcal{I}$ in increasing o-index order as $k_1 \triangleleft k_2 \triangleleft \cdots$ (or $k_1 \triangleleft \cdots \triangleleft k_n$) and write $D_m := \{k_1, \ldots, k_m\}$. Applying Lemma S2.2 to the marginal at k_1 gives a measurable map $g_1 : [0,1] \to \mathcal{X}_{k_1}$ reconstructing X_{k_1}, and applying Lemma S2.2 inductively to the kernel $Q^{\nu}_{k_m}$ on $\mathcal{X}_{D_{m-1}} \times \mathcal{B}_{k_m}$

gives measurable maps $g_m : \mathcal{X}_{D_{m-1}} \times [0,1] \to \mathcal{X}_{k_m}$ for $m \geq 2$. We then define $G_o : [0,1]^{\mathcal{I}} \to \mathcal{X}_{\mathcal{I}}$ coordinatewise by

$$[G_o(u)]_{k_1} := g_1(u_{k_1}), \qquad [G_o(u)]_{k_m} := g_m\big(G_o(u)_{D_{m-1}}, u_{k_m}\big) \quad (m \geq 2).$$

By induction, each coordinate map is measurable, and hence G_o is measurable into the product space $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}})$; note that G_o depends only on ν and o.

Under $\nu \otimes \lambda^{\mathcal{I}}$, induction on m via Lemma S2.2 shows that $G_o(T_o(X,V))_{D_m} = X_{D_m}$ a.s. for all m, and taking a countable union over m gives Equation (4) under $\nu \otimes \lambda^{\mathcal{I}}$. For any $\mu \ll \nu$, we have $\mu \otimes \lambda^{\mathcal{I}} \ll \nu \otimes \lambda^{\mathcal{I}}$, so Equation (4) holds under $\mu \otimes \lambda^{\mathcal{I}}$ as well, and pushing forward $T_o(X,V) \sim \rho_o^\mu$ by G_o yields $(G_o)_{\#}\rho_o^\mu = \mu$. □

Proof of Theorem 4.2.

Proof. For every $\mu \in \mathcal{M}$, we have $\rho_o^\mu = \mu \mathsf{P}_o$ by definition and $\rho_o^\mu \mathsf{R}_o = \mu$ by Equation (5), where neither Markov kernel P_o nor R_o depends on μ. By the randomization criterion (Blackwell, 1953; Torgersen, 1991), the original and transformed experiments are equivalent. Similarly, for any $o, o' \in \mathfrak{O}$,

$$\rho_o^\mu(\mathsf{R}_o \mathsf{P}_{o'}) = (\rho_o^\mu \mathsf{R}_o)\mathsf{P}_{o'} = \mu \mathsf{P}_{o'} = \rho_{o'}^\mu$$

and $\rho_{o'}^\mu(\mathsf{R}_{o'} \mathsf{P}_o) = \rho_o^\mu$, so the μ-independent composite kernels $\mathsf{R}_o \mathsf{P}_{o'}$ and $\mathsf{R}_{o'} \mathsf{P}_o$ establish equivalence between any two transformed experiments. □

Proof of Corollary 4.3.

Proof. Because P_o is a Markov kernel and $G_{o'}$ is measurable, $\psi' = (\mathsf{P}_o \psi) \circ G_{o'}$ is a $[0,1]$-valued measurable test, and Lemma 4.1 ($(G_{o'})_{\#}\rho_{o'}^\mu = \mu$) gives

$$\int \psi' \, \mathrm{d}\rho_{o'}^\mu = \int \mathsf{P}_o \psi \, \mathrm{d}\mu = \int \psi \, \mathrm{d}(\mu \mathsf{P}_o) = \int \psi \, \mathrm{d}\rho_o^\mu \qquad \text{for all } \mu \ll \nu.$$

Moreover, if $U' \sim \rho_{o'}^\mu$ and $V \sim \lambda^{\mathcal{I}}$ is drawn independently, then $G_{o'}(U') \sim \mu$ and hence $T_o(G_{o'}(U'), V) \sim \rho_o^\mu$; conditional on $U' = u'$, the randomized draw $T_o(G_{o'}(u'), V)$ rejects under ψ with probability $\psi'(u')$ by Equation (6). □

Proof of Proposition 4.5.

Proof. For any measurable $h : [0,1]^{\mathcal{I}} \to [0,\infty]$, applying Equation (4) under $\nu \otimes \lambda^{\mathcal{I}}$ and using $\rho_o^{\nu} = \lambda^{\mathcal{I}}$ gives

$$\begin{aligned}\int h(u)\,\rho_o^{\mu}(\mathrm{d}u) &= \int h(T_o(x,v))R(x)\,(\nu \otimes \lambda^{\mathcal{I}})(\mathrm{d}(x,v)) \\ &= \int h(T_o(x,v))R(G_o(T_o(x,v)))\,(\nu \otimes \lambda^{\mathcal{I}})(\mathrm{d}(x,v)) \\ &= \int h(u)R(G_o(u))\,\lambda^{\mathcal{I}}(\mathrm{d}u),\end{aligned}$$

from which $\rho_o^{\mu} \ll \lambda^{\mathcal{I}}$ and Equation (8) follow by taking h to be indicator functions. □

Proof of Corollary 4.6.

Proof. Because P_o maps (μ,ν) to $(\rho_o^{\mu}, \lambda^{\mathcal{I}})$ and R_o maps $(\rho_o^{\mu}, \lambda^{\mathcal{I}})$ back to (μ,ν) by Equation (5), applying the data-processing inequality (Liese, 2012, Theorem 3.1) in both directions gives

$$D_f(\mu\|\nu) \le D_f\big(\rho_o^{\mu}\big\|\lambda^{\mathcal{I}}\big) \le D_f(\mu\|\nu).$$

□

Proof of Corollary 4.7.

Proof. Write $o = (\vec{\preceq}, \trianglelefteq)$ and let $T_D := T^{\vec{\preceq},\trianglelefteq}_{D,\nu_D}$ be the finite-prefix EGRT of ν_D defined prior to the proof of Theorem 2.6. By Proposition S2.4, $\pi_D^{[0,1]} \circ T_o = T_D \circ (\pi_D^{\mathcal{X}} \times \pi_D^{[0,1]})$ pointwise, so

$$\rho^{\mu}_{o,D} = \big(\pi_D^{[0,1]}\big)_{\#}\rho_o^{\mu} = (T_D)_{\#}(\mu_D \otimes \lambda^D).$$

Since $\mu \ll \nu$ implies $\mu_D \ll \nu_D$, applying Corollary 4.6 to T_D on the finite index set D yields Equation (10). □

Proof of Corollary 4.8.

Proof. By Corollary 4.7, $D_f(\rho^{\mu}_{o,D_m}\|\lambda^{D_m}) = D_f(\mu_{D_m}\|\nu_{D_m})$. Since (μ_{D_m},ν_{D_m}) is a coordinate projection of both $(\mu_{D_{m+1}},\nu_{D_{m+1}})$ and (μ,ν), the data-processing inequality (Liese, 2012, Theorem 3.1) implies that $D_f(\mu_{D_m}\|\nu_{D_m})$ is nondecreasing in m and bounded above by $D_f(\mu\|\nu)$.

For the reverse inequality, work on $(\mathcal{X}_{\mathcal{I}}, \mathcal{B}_{\mathcal{I}}, \nu)$, put $\mathcal{F}_m := \sigma(\pi_{D_m})$, and let $R := \mathrm{d}\mu/\mathrm{d}\nu$. Then $R_m := \mathbb{E}_{\nu}[R \mid \mathcal{F}_m]$ is a version of $(\mathrm{d}\mu_{D_m}/\mathrm{d}\nu_{D_m})(X_{D_m})$, and since $D_m \uparrow \mathcal{I}$ implies $\bigvee_m \mathcal{F}_m = \mathcal{B}_{\mathcal{I}}$, Lévy's upward martingale convergence theorem (Kallenberg, 2021, Chapter 9) gives $R_m \to R$ ν-a.s. Choosing a subgradient

$a \in \mathbb{R}$ of f at 1, the nonnegative convex function $\widetilde{f}(t) := f(t) - a(t-1)$ satisfies

$$D_f(\mu_{D_m} \| \nu_{D_m}) = \mathbb{E}_\nu[\widetilde{f}(R_m)], \qquad D_f(\mu \| \nu) = \mathbb{E}_\nu[\widetilde{f}(R)].$$

By continuity of $\widetilde{f}$ on $[0, \infty)$ (with $\widetilde{f}(0) = \lim_{t \downarrow 0} \widetilde{f}(t) \in [0, \infty]$), we have $\widetilde{f}(R_m) \to \widetilde{f}(R)$ ν-a.s., and Fatou's lemma yields $\liminf_{m \to \infty} D_f(\mu_{D_m} \| \nu_{D_m}) \geq D_f(\mu \| \nu)$. $\square$